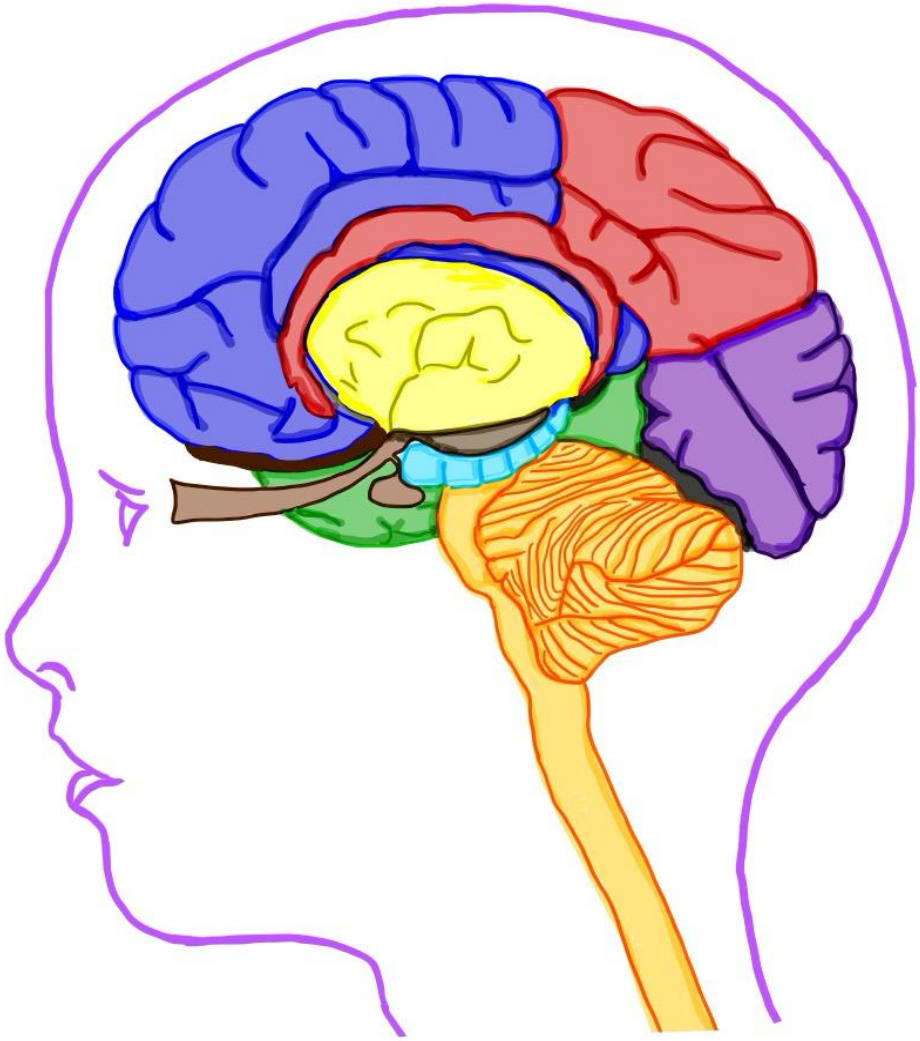
PEDIATRIC

Central nervous system



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Seizures (Convulsions)

Seizure: Transient occurrence of symptoms &/or signs due to abnormal brain electrical activity manifested as **motor** activity, sensory disturbance, disturbed **consciousness, autonomic, psychic** or **mixed** manifestations.

Convulsions: Involuntary muscular contraction (**motor**)due to abnormal electrical activity.

- **Causes of childhood convulsions**

Convulsions with fever	Convulsions without fever
1. **Febrile convulsions** 2. Meningitis 3. Encephalitis 4. Brain abscess.	1. Epilepsy 2. Electrolytes disturbance (low Ca –low Sugar –low mg – low & high Na) 3. Intracranial hemorrhage (head trauma) 4. Space occupying lesion (brain tumors) 5. Drugs ingestion or toxicity

Note: **Febrile convulsion is the most common cause of convulsion in children.**

Febrile Convulsions

- **Definition:** Is a seizure associated with fever in age group from 6 months to 6 years without any CNS Infection (CNS FREE)

  (The Focus is outside CNS AS URTI, UTI, GE, pneumonia (**Usually viral URTI infection > 85% of cases**)

- **Epidemiology:**
- **Incidence:**

  2– 4% of children.
  Genetic predisposition 10 – 20%
  Sex Slightly more in boys
  Recurrent 30 – 40% of cases

- **Risk Factors for Recurrence of Febrile Seizures:**

  MAJOR
  1. Age < 1 yr younger age)
  2. Duration of fever < 24 hr (shorter the duration of illness before the seizure).
  3. Fever 38–39°C (the lower the temperature at the time of seizure).

  MINOR
  1. Family history of febrile seizures
  2. Family history of epilepsy
  3. Complex febrile seizure
  4. Daycare
  5. Male gender
  6. Lower serum sodium at time of presentation

- **pathophysiology:** **is unknown**
  (Genetic predisposition + Fever (The seizure usually occurs early in a viral infection when the temperature is rising rapidly)

- **Types:** مهم جدا كلينك

	Simple febrile seizures	Complex febrile seizures
incidence	more common 85 %	less common
Type	Generalized tonic–clonic	Focal or localized
Duration	< 15 minutes	> 15 minutes
Frequency	Only one fit during 24 hr	Recurrent
Postictal neurological deficits	No	Mybe

Simple febrile seizures: do **not** cause **brain damage**/ the child's subsequent **intellectual performance** is the same as in children who did not experience a febrile seizure. There is a **1%–2% chance** of subsequently developing an epilepsy, **similar to the risk for all children.**

- **Management of (Fever + convulsion)**
  The child presentation may be:
-

1. in convulsion (Emergency)

ABC
Airways: put child on Left Lateral position –Clear airwags (remove ang foreign body & do (suction
.connect patient to monitor.
.Breathing: high fLow oxygen
Circulation : check pulse – insert IV Line (canula)– serum sugar & Ca
If ca or sugar is Low corrected immediately with D10% or Ca gluconate)
Once you establish IV access (canula)
→ Start anticonvulsant medication to stop convulsive activity:
Diazepam 0.3–0.5 mg /kg /dose or
Lorazepam 0.1mg /kg /dose. IV state
→.If no response or recurrent repeat it twice
If the child still in convulsions or recurrent
Now start either [phengtoin or phenobarbital Loading & maintenance dose
:If still in convulsion state:
Anaesthetist MUST be present Rapid Sequence Induction of anaesthesia with thiopental (thiopentone)..ETT &MV
In no IV access (no canula)
Give buccal madapolam
OR Rectal diazepam
OR rectal para aldehyde

2. Rx of child presented in post ictal state:

1. good history (preictal – Ictal – postictal)

2. Examination – General (Anterior Fontanelle.)
   (Cause of Fever)? + CNS examination
   + Sign of meningeal (If the age > 18 months.)

3. investigations

1. CBC, CRP, serum sugar & Ca, electrolytes
2. If the child age < 18 months → Lumbar puncture should be Done to exclude meningitis (B/C in these age AF still open & symptoms & signs of meningitis are nonspecific)
3. If the child age > 18 months → look for meningeal signs
   - Meningeal sign (+ve) → (L.P Should be done)
   - Meningeal sign (−ve) → (no need to do LP & deal with pt case of F. Convulsion)
4. CT scan brain & EEG are not indicated in case of simple febrile convulsions.

- **Treatment of febrile convulsion:**
  - only Supportive in most cases (admission – good hydration – Antipyretic (paracetamol).
  - Treatment of Underlying cause (focus) according to your assessment for example if URTI only hydration & antipyretic – if UTI gives antibiotics.
  - Parents need reassurance & advice sheets.
  - Antipyretics are given but have not been shown to prevent febrile seizures.
  - Oral prophylactic anticonvulsant drugs are not used, as they do not reliably reduce the recurrence rate of seizures, and have a relatively high risk of adverse effects.

The risk of epilepsy in future increased in:

1. Complex type of febrile convulsion (4–12%) (Those which are focal, prolonged, or repeated in the same illness)
2. child +ve F/H of epilepsy.
3. Abnormal neurologic signs or development.

Epilepsy

- **Epilepsy:** is Recurrent (=>2) unprovoked seizures. (0.05% incidence)
- **Status epilepticus:** Its prolonged seizure lasting more than 30 minutes or repetitive seizures without regaining consciousness. (new definition >5 min).
- **Epilepsy syndromes:** it's seizures that are associated with characteristic EEG Findings or additional clinical or pathological Features.

- **Epidemiology:** epilepsy incidence of about 0.05% (less common during first year of life) and a prevalence of 0.5% (Affects 1 in 200 children)

- **Types of Epilepsy:**

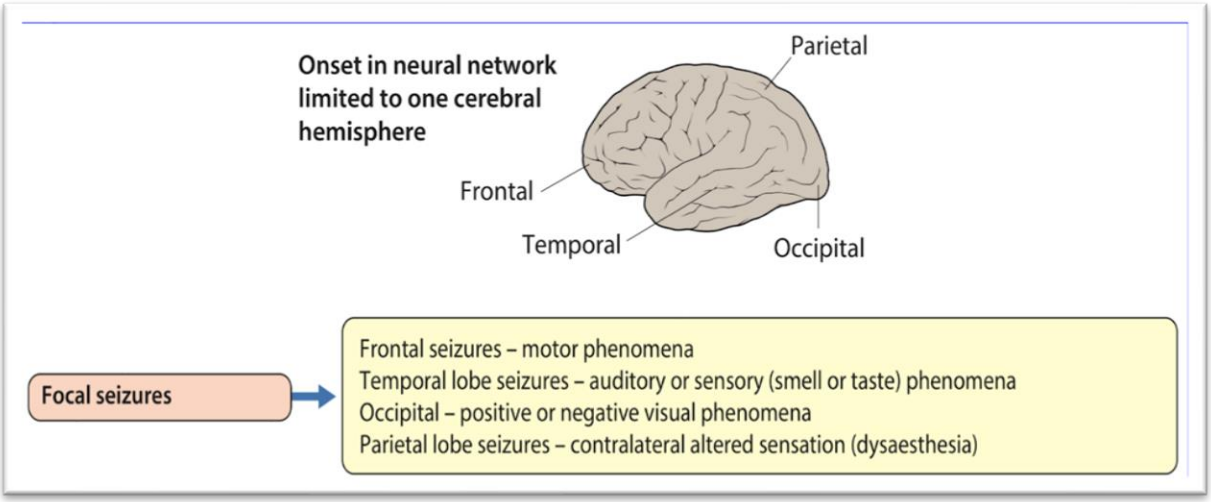
Focal (partial)seizures (where seizures arise from one or part of one hemisphere)	Generalized seizures (discharge arises from both hemispheres).	Epilepsy syndrome
• Simple partial seizures: (Intact consciousness)	• Tonic clonic (Grand mal)	• Infantile spasms.
• Complex partial seizures: (Impaired consciousness)	• Absence (Petit mal)	• Juvenile myoclonic epilepsy
• Partial with 2ry generalization	• Myoclonic seizures.	• Benign Rolandic epilepsy
//	• Astatic (Atonic)	//

- **Causes of epilepsy:**

1. Idiopathic (1ry)	2. Secondary (20%)
Genetic (70%–80%) – also called (idiopathic), caused by alleles at several loci together rather than a single gene, so inheritance is (complex)	• Malformation/Cerebral dysgenesis. • Post brain damage: ischemia /HIE / infarction/congenital infection • Post traumatic (IC Hge) • Neurocutaneous syndromes, e.g. tuberous sclerosis • Post meningitis. • Cerebral tumor/Neurodegenerative disorders

Focal seizures

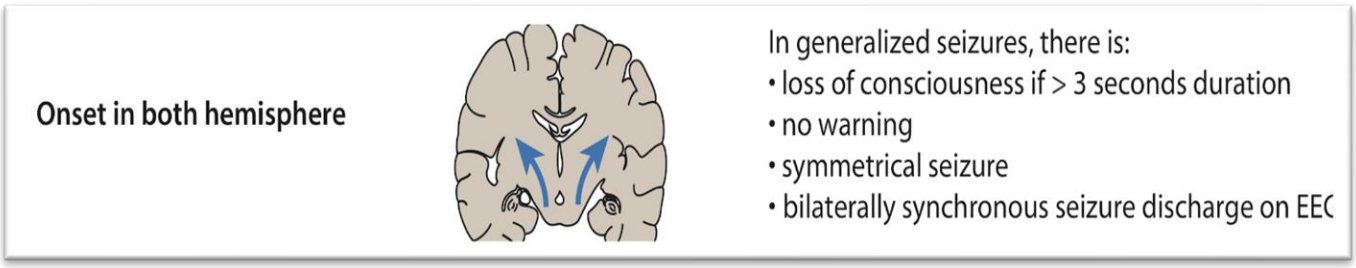
- originate in a relatively small group of dysfunctional neurons in one of the cerebral hemispheres
- may be heralded by an **aura** (the sensory epileptic seizure) which reflects the site of origin
- may or may not be associated with change in **consciousness** or evolve to generalized tonic–clonic seizure



1. Simple partial seizures: (consciousness not impaired)
   *motor signs.
   *Special sensory (visual – auditory – olfactory)
   *Autonomic.
2. Complex partial seizures: (Consciousness impaired)
3. Partial seizures with secondary generalization: Focal seizures followed by generalized tonic clonic seizure.

- Treatment of partial seizures: Carbamazepine (D.O.C) Tegretol

Generalized seizures



1. Tonic clonic seizures: (Grand mal epilepsy)

✓ is the commonest type of generalized seizures.
it passes through:

- **ictal phase:** Rhythmical contraction of muscle groups following by tonic phase. Usually associated with: (1) up rolling of eyes.
  (2) Frothy secretion from mouth.
  (3) cyanosis
  (4) Biting of the tongue.
  (5) incontinence of urine.
  (6) Loss of consciousness
- **Post ictal phase:** usually lasts from a few seconds to minutes, followed by unconsciousness or deep sleep for up to several hours.

- **Treatment of Tonic clonic seizures:** **Na valproate** (D.O.C) **depakin**

2. **Absence seizures (petit mal)** 2%

Sudden brief **Loss** of **consciousness** without Loss of postural tone.

- It may be associated with automatisms such as Lip smacking, Flickering of eyelids.
- **abrupt** onset and termination (Duration 5–30 seconds).
- Child has **no recall** except realizes they have missed something.
- **No aura – No post-ictal stage.**
- It may be triggered by **hyperventilation.**
- Attack can be **recurrent** many times / Day
- **2 types** (**typical –Atypical**)
- Sex: more common in **female** (⅔ of cases)
- Age: 4–12 years of age (**school age**)
- **Developmentally normal** but can interfere with schooling.
- EEG shows fast generalized spike and **wave** (**3–4 Hz**) **discharges**, bilaterally synchronous
- Treatment: **Ethosuximide** (D.O.C)
- Prognosis **good with 80% remission in adolescence/ a** few evolve into juvenile absence or juvenile myoclonic epilepsy.

3. **Myoclonic seizures**: Brief, often repetitive, jerking movements of the limbs, neck or trunk.

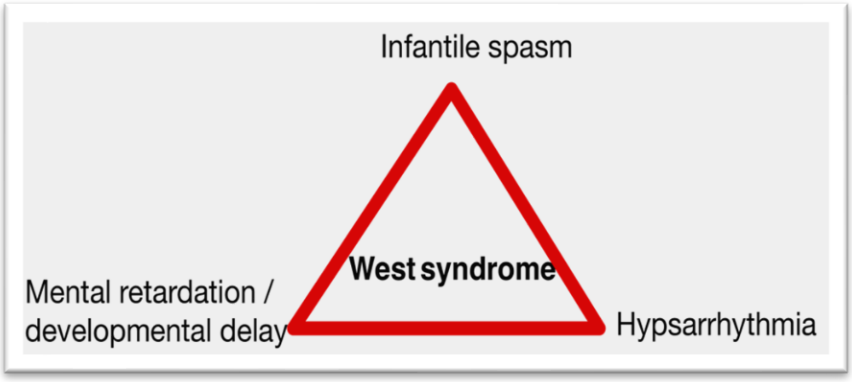
4. **Atonic seizures**: a transient loss of muscle tone causing a sudden fall to the floor or drop of the head, Often combined with a myoclonic jerk.

Epilepsy syndromes

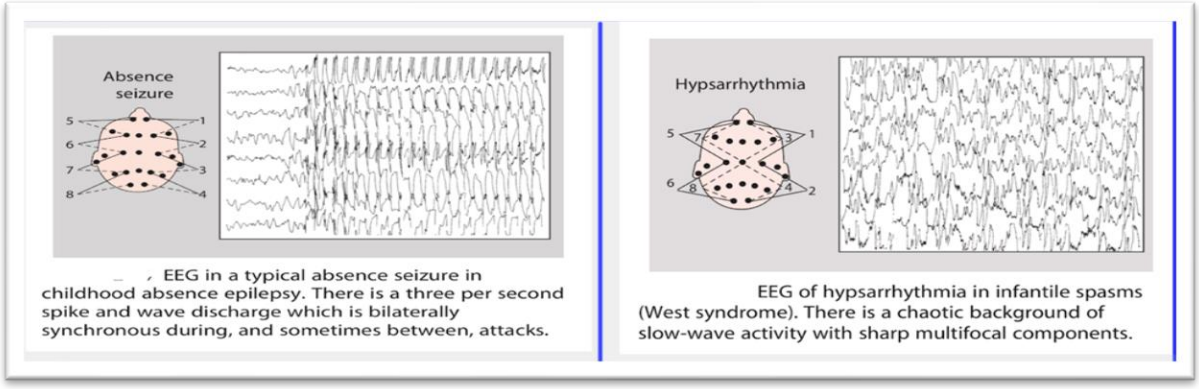
Infantil spasm (west syndrome)

- are brief contraction of the neck, trunk, and limbs (Flexor muscle) Followed by extension of the arms (Salaam Spasm)
- it's **repetitive** 20–30 & last 1–2 seconds.
- usually occur when the child is waking from or going to **sleep.**
- age: **3 – 12 months** (>85 % of cases < 1 year)
- Sex: more in **boys**
- It may be **misdiagnosed as infantile colic**
- Causes:
  1. **Idiopathic** 20 –30%
  2. **2ry to neurological cause** 70 – 80%
     A. **Tuberous sclerosis** (hypopigmentation +seizures)
     B. Birth asphyxia
     C. Inborn error of metabolism.

- EEG: shows **Hypsarrhythmia** (Chaotic pattern of high voltage slow waves)
- Treatment: (1) **Vigabatrin** (2) **ACTH.** (3) Corticosteroids.
- Prognosis:
  (1) Idiopathic type: **better** response to treatment & **good** prognosis.
  (2) Symptomatic type: **poor** response to treatment & **poor** prognosis



	Absence seizures (petit mal)	Infantile spasm (west syndrome)
• Age	School age (4–12 years)	Infants (4–12 months)
• Sex	More in girls	More in boys
• Cause	Unknown	1ry or 2ry to neurological diseases
• EEG	3 spikes &waves /second	Hypsarrhythmia
• DOC	Ethosuximide	Vigabatrin in 2ry, ACTH OR Cortisone in 1ry
• Prognosis	Good	Variable if 2ry bad prognosis (MR, D delay)



Benign focal epilepsy (Epilepsy with centro–temporal spikes or Rolandic epilepsy) mcq

- **Tonic–clonic seizures** in sleep, or simple **focal** seizures with **awareness** of abnormal feelings in **the tongue and distortion of the face** (supplied by the **Rolandic** (centro–temporal) **area** of the brain).
- Conscious usually intact
- Age: **4 – 10 yr.**
- % : **15%** of all childhood epilepsy.
- +ve F/H in about 15% of cases.
- ECG: shows **centrotemporal** spikes

- Treatment: usually responds to anti-convulsant drugs
- Prognosis: it's benign and may not require AEDs. Remits in adolescence

Lennox – Gastaut syndrome:

- Age: 1 – 3 yr
- Multiple seizure types (atonic – tonic – atypical)
- maily children have underlying brain injury or malformation
- Respond poorly to Anticonvulsant Rx.
- Associated with neurodevelopmental arrest or regression and behaviour disorder.

Juvenile myoclonic epilepsy:

- Age: 10–20 years
- Myoclonic seizures, generalized tonic–clonic seizures, and absences may occur, mostly shortly after waking.
- Learning is unimpaired.
- Characteristic EEG.
- Response to treatment is usually good but lifelong.
- Remission is unlikely.

- Diagnosis: The diagnosis of an epilepsy is primarily based on a detailed history from the child and eyewitness + investigation:

1. EEG: **is Investigation of choice**, normal EEG doesn't exclude epilepsy.
   24 hr ambulatory EEG (may needed)
2. Imaging **(CT scan Brain – MRI Brain)**
3. Other investigations

   Metabolic investigations will be indicated if there is developmental arrest or regression, or seizures are related to feeds or fasting.
   Genetic tests are becoming increasingly useful, especially in intractable epilepsies with developmental arrest or delay ('epileptic and developmental encephalopathies').

- Management of epilepsy:

1–Education:

- Parent education about aetiology, medications (dose & side effects), prognosis.
- Regular follow up is essential.
- Jumping from one doctor to another is not recommended.
- Don't stop antiepileptic drugs suddenly.
- Ovoid triggering factors & as video games
- No restriction of physical activity but some activities should be attended (swimming…)

2–Antiepileptic drug therapy (AEDs):

- Not all children with epileptic seizures require antiepileptic drug (AED) therapy.
- Choose an appropriate AED (DOC) for the seizure and epilepsy.
- Monotherapy at the minimum dosage to prevent the seizures without adverse effects is the desired goal, although in practice more than one AED may be required.
- All AEDs have potential unwanted effects and these should be discussed with the child and parent.
- AED therapy may be discontinued after 2 years free of seizures, but should usually be continued indefinitely in young people with juvenile absence epilepsy or juvenile myoclonic epilepsy.

Choice of anti-epileptic Drugs. (NICE)

Seizure type	First Line مهم	Second Line غير مهم
• Tonic – Clonic • Absence seizures • Myoclonic seizures • Focal epilepsies	• Valproate – carbamazepine • Ethosuximide, valproate • Valproate, levetiracetam • Carbamazepine, lamotrigine	• Clobazam, levetiracetam. • Clobazam, levetiracetam. • Clobazam • Valproate.

Table 29.3 Common or important unwanted effects of antiepileptic drugs

Drug	Adverse-effects
Valproate	Weight gain, hair loss, teratogenic, rare idiosyncratic liver failure
Carbamazepine	Rash, ataxia, liver enzyme induction can reduce effect of oral contraception
Lamotrigine	Rash, insomnia, ataxia
Ethosuximide	Nausea and vomiting
Levetiracetam	Irritability
Gabapentin	Insomnia
Topiramate	Weight loss, depression, paraesthesia
Vigabatrin	Irritability, visual field restriction limits its use mainly to infantile spasms

All the above drugs may cause drowsiness.

Other treatment options in children with intractable epilepsies,

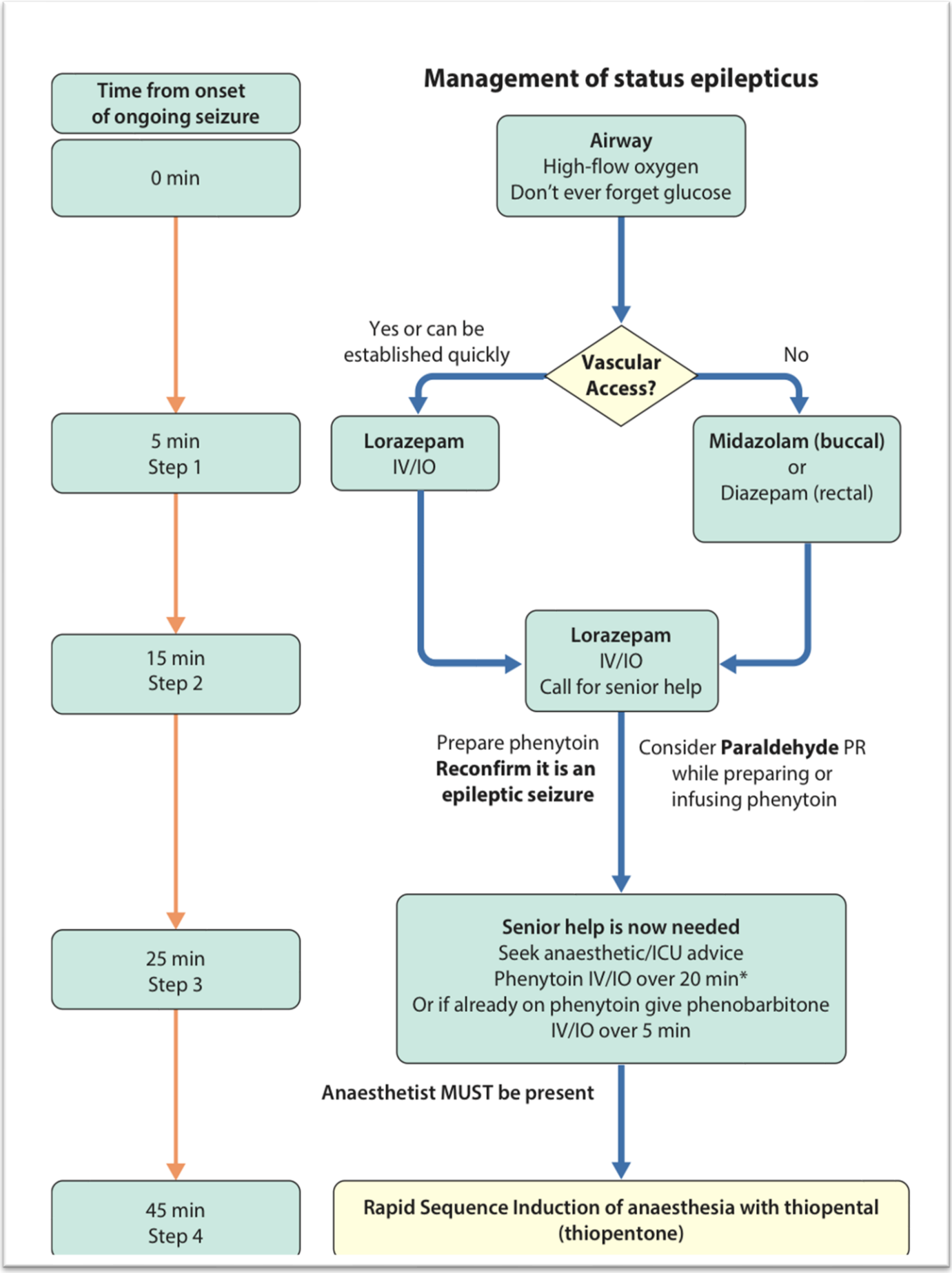
- **Ketogenic** (low-carb, fat-based) diets may be helpful in some children.
- **Vagal nerve stimulation**
- **Epilepsy surgery.** The main procedure is temporal lobectomy for mesial temporal sclerosis, but other procedures include hemispherectomy (disconnection of the hemisphere) and other focal resections.

Status epilepticus

Definition: Established status epilepticus is an epileptic seizure lasting 30 minutes or repeated epileptic seizures for 30 minutes without recovery of consciousness (new definition 5 minutes or more)

Refractory_status_epilepticus (RSE)
Clinical or electrographic seizures which persist after an adequate dose of an initial benzodiazepine and a second appropriate anticonvulsant drugs.

Super_refractory_status_epilepticus (super-RSE)
That persists or recurs following the administration of continuous IV infusion/ anesthetics for more than 24 h.


Management of status epilepticus
Time from onset of ongoing seizure
0 min
5 min Step 1
15 min Step 2
25 min Step 3
45 min Step 4
Airway
High-flow oxygen
Don't ever forget glucose
Vascular Access?
Yes or can be established quickly
No
Lorazepam IV/IO
Midazolam (buccal) or Diazepam (rectal)
Lorazepam IV/IO Call for senior help
Prepare phenytoin Reconfirm it is an epileptic seizure
Consider Paraldehyde PR while preparing or infusing phenytoin
Senior help is now needed
Seek anaesthetic/ICU advice
Phenytoin IV/IO over 20 min*
Or if already on phenytoin give phenobarbitone IV/IO over 5 min
Anaesthetist MUST be present
Rapid Sequence Induction of anaesthesia with thiopental (thiopentone)

Conditions that Mimic Seizures / Paroxysmal disorders (Funny turns)

1–Breath Holding attacks (Spells) BHA التفحيمة mcq

Cyanotic BHA (Common)	Pallid BHA (Less common)
• more common • Age: toddler (6 months –5years) • Mechanism: toddlers when they are upset —>The child cries, holds his breath in expiration and goes blue & Sometimes children will briefly lose consciousness but rapidly recover fully. • EEG: normal • Prognosis: Self–limited, outgrown within a few years (at 5 years) • Treatment: Reassurance (behavior modification therapy with distraction, may help) • Iron therapy of the coexisting iron deficiency may be helpful.	• less common • Age: toddler (6 months –5years) • Mechanism: The spell is always provoked by • painful stimulus: (minor Head trauma) →Reflex vagal–cardiac bradycardia (goes pale & may briefly lose consciousness but rapidly recover fully. • EEG: normal • Prognosis: Self–limited, outgrown within a few years (at 5 years) • Treatment: Reassurance (behavior modification therapy with distraction, may help) • Iron therapy of the coexisting iron deficiency may be helpful.

2– Syncope (fainting) vasovagal:

Children may faint if in a hot and stuffy environment, on standing for long periods, or from fear.
Clonic movements lasting a few seconds are common.
Self–limiting condition

3–Benign Paroxysmal vertigo of childhood:

This is characterized by recurrent attack of vertigo Lasting from one to several minutes, associated with nystagmus, dizziness, or even falling, normal CNS examination, no treatment
(B/C attack is brief & disappears spontaneously)

4–Psevdo–Seizures:

Common in girls, > 10 yr
Seizure not typical, No cyanosis, No incontinence, No injury, No tongue biting.
EEG: Normal. Diagnosis by exclusion, needs psychotherapy.

5– Infantile masturbation or child Gratification disorder:

Age & sex: Females (2 months –3years), Idiopathic.
C/P: Attacks of tonic posturing with copulatory movements followed by flushing & perspiration.
Normal neurological examination, No loss of consciousness, Normal EEG
Rx: Spontaneous resolution at 3 yrs

6–Spasmus nutans: Triad of nystagmus, head tilt (Torticollis), and head nodding
- Brain MRI & EEG: Normal
- Prognosis: Disappear before 5 yr of age

- Others conditions mimic Epilepsy:
  (1) Benign Paroxysmal torticollis.
  (2) Cyclic Vomiting syndrome
  (3) Night manes – Night terrors – Tics
  (4) Reflex anoxic syncope.
  (5) Cardiac Arrhythmia

Meningitis

- **Definition:** Inflammation of the meninges covering the brain.

- **Types & Etiology:**

Bacterial meningitis = (septic / pyogenic)

It's a serious infection in children (MR = 5 – 10% & ≈10% of survivors develop long term neurological impairment.

According to age:

<3 months	1 month to 6yrs	>6yrs
• Group B strep • Escherichia coli • Listeria monocytogenes	• Neisseria meningitidis • Strep pneumoniae • Hib	• Neisseria meningitidis • Strep pneumoniae

Viral meningitis = Aseptic

Viral infections are the **most common** cause & usually is **self-limiting**.

Causes as

1. Enterovirus as coxsackieviruses
2. Mumps virus.
3. Adenovirus
4. Epstein _ Barr virus.
5. HIV.

Other causes of meningitis (Rare)

1. **T.B Meningitis** is rare, usually in children <5 years

2. In immunodeficiency child:

Parasites: as Toxo.
Or
Fungi: as Histoplasma.

Risk Factors:

1. Poverty & overcrowding.
2. Lack of specific immunization
3. Meningomyelocele – meningocele.
4. Ventriculoperitoneal shunt: V-P shunt (Staph epidermidis).
5. Immunodeficiency:
   - Complement (predispose for meningococci).
   - Asplenia (congenital or traumatic) or splenic dysfunction, auto splenectomy (e.g. sickle cell disease).
   - T-cell defects (congenital or acquired by chemotherapy, AIDS and malignancy)
6. more in Male / Black

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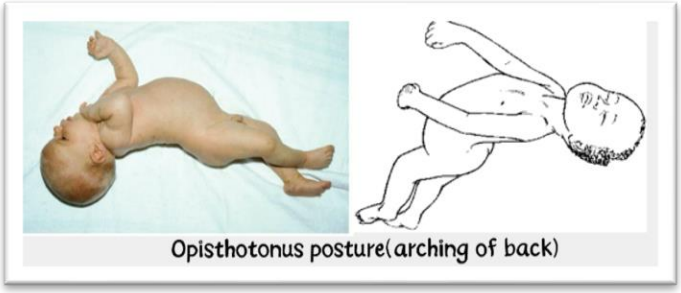
Route of transmission:

- **Droplet infection (most common)**
- Direct (Otitis media – orbital cellulitis)
- Hematogenous. (through **Blood**) (in **Neonate**)

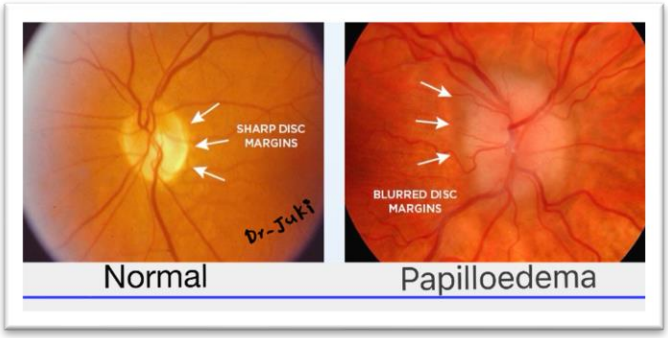
Clinical picture:

Infants& young children (AF OPEN)	Older children
Systemic manifestation (Non specific) Fever – poor feeding– Irritability vomiting– diarrhea –Lethargy Mottled skin –Petechial rash in meningococcal. **May also presented with this sign:** Bulging & Tense anterior Fontanelle.	**Symptoms & signs become more specific** **General symptoms:** Fever –irritability–poor feeding –ill – fatigability. **Symptoms of increased ICP:** Projectile vomiting

Arched back (opisthotonus)



Papilledema is (rare –very late sign)



Symptoms of neurological deficits

Drowsiness
Loss of consciousness (coma)
Seizures
Cranial nerve palsy (6th cranial nerve)

Headache
Photophobia, blurred vision.

Later: **Cushing triad:**
1. hypertension 2. Bradycardia
3. irregular breathing) → loss of consciousness.

Symptoms of neurological deficits

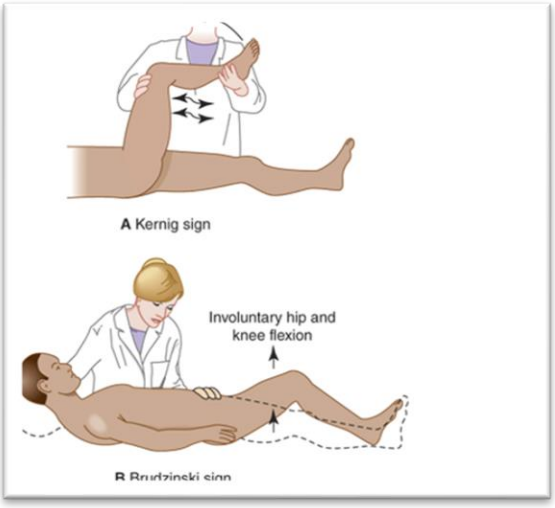
- Drowsiness
- Loss of consciousness (coma)
- Seizures
- Cranial nerve palsy (6th cranial nerve)
- Focal neurological signs / paralysis

Signs of meningeal irritation:

1–Neck stiffness –neck pain & nuchal rigidity on passive movement. (**seen also in** severe tonsillitis –upper lobar pneumonia –cervical lymphadenopathy)

2–Kernig's sign: Position the patients supine with their hips & knees flexed to 90°. This test is positive if there is pain on passive extension of the knee.

3–Brudzinski's sign: Position the patients supine and passively flex their neck. This test is positive if this maneuver causes reflex flexion of the both hip and knee.



Investigations:

1. CBC: WBC count with differential (mainly neutrophils in bacterial)
2. CRP Titre: high
3. Urea / electrolyte– blood glucose.
4. Rapid antigen test for meningitis organisms (can be done on blood, CSF, or urine)

Samples for viral PCRs (e.g. throat swab, nasopharyngeal aspirate, conjunctival swab, stool sample).

5. **Lumbar puncture:** CSF examination:
   For (diagnosis of meningitis– identity the causative organism – Determine appropriate antibiotic – Evaluate the treatment).

Absolute Contraindications of lumbar puncture:

1. Cardiorespiratory instability (as shock –cardiac arrest).
2. Signs of raised intracranial pressure e.g. coma, high BP, low heart rate or papilledema → risk of coning & herniation of the cerebellum through the foramen magnum → sudden arrest.
3. Coagulopathy & Thrombocytopenia
4. Focal neurological signs
5. Local infection at the site of LP

- If one of the above contraindications is present don't delay starting antibiotics.



- Site of LP: (L3–L4) or (L4–L5).

CSF	Normal	Bacterial meningitis	Viral meningitis	TB meningitis
Appearance (color)	Clear	Turbid	Clear	Turbid/clear/ viscous
White blood cells	0–5/mm3 (in newborn up to 30)	high mainly Polymorphs: ↑↑↑	high mainly Lymphocytes: ↑	high mainly Lymphocytes: ↑
Protein	20–40 mg/dl or 0.15–0.4 g/L	high ↑↑	Normal or high ↑	high ↑↑↑
Sugar	≥50–60 % of blood sugar or ≥ ⅔.	low ↓↓	Normal (Low in enterovirus –mumps)	low ↓↓↓
Gram stain	Negative	positive	negative	TB bacilli ZN stain
Culture & sensitivity	Negative	positive in most of cases	negative	negative

Partially treated bacterial meningitis مهمة

Children are frequently given oral antibiotics for a nonspecific febrile illness. If they have early meningitis, this partial treatment with antibiotics may cause diagnostic problems. CSF examination shows a markedly raised number of white cells, but cultures are usually negative. Rapid antigen screens and PCR are helpful in these circumstances. Where the diagnosis is suspected clinically, a full course of antibiotics should be given.

Treatment of bacterial meningitis:

Specific Rx (Antibiotic IV)	Supportive Rx
Start empirical antibiotic (depend on common pathogens) 1. Age < 3 months: • Ampicillin + gentamicin (1st line) • Ampicillin + 3rd generation cephalosporin. (Cefotaxime or ceftriaxone) 2. Age > 3 months: • 3rd generation cephalosporin **Route:** Intravenous only **Duration:** 10– 14 days. (In neonate 2–3 wks.)	1. Antipyretic for fever. 2. IVF (⅔ M) why? To ovoid SIADH 3. Dexamethasone: 0.6mg / kg/day every 6hrs IV for 2 Day, given 1/2 hr before antibiotic, to reduce the risk of long–term complication (deafness). 4. Anti–convulsant (if there is convulsions) 5. Observation. (Vital sign: temp – OFC daily – Wt daily – input & output chart 6. Treatment of complications.

- **Prognosis of meningitis depend on: Age** (Younger age → worse prognosis) –/ **type** of organism / **Early** diagnosis & early treatment → good prognosis)

- **Complications of meningitis:**

Early	Late
1. Shock: usually is septic. 2. Subdural effusion: confirmed by cranial CT or MRI scan. Most resolve spontaneously, but some require neurosurgical intervention, associated with Hib & pneumococci. 3. Brain abscess: children become more toxic, ill, high–grade fever & sometimes associated with focal signs & seizures. It is confirmed on cranial CT or MRI scan. Drainage of the abscess is required. 4. Cranial nerve palsy due to local vasculitis. 5. SIADH (syndrome of inappropriate ADH secretion): prevention by IVF restriction. 6. DIC & Waterhouse Friderichsen syndrome. 7. Hydrocephalus – may result from impaired resorption of CSF (communicating hydrocephalus) or blockage of the cerebral aqueduct or ventricular noncommunicating hydrocephalus). A ventricular shunt may be required. 8. Cerebral edema & local cerebral infarction – this may result in focal or multifocal seizures.	1. Epilepsy 2. Hearing loss (SNHL) ~~أصم~~: is the most common long–term sequelae of meningitis, due to inflammatory damage to the cochlear hair cell, associated with strep pneumonia & Hib meningitis. 3. Vision & speech disturbances 4. MR & Learning disability 5. Cerebral palsy.

- **Prophylaxis**
  - **Rifampicin** or ciprofloxacin to eradicate nasopharyngeal carriage is given to all household contacts for meningococcal meningitis and Hib infection.
  - Prevention with **vaccination** (**Hib vaccine – Meningococcal vaccine ACYW135 –pneumococcal vaccine**).

Meningococcal infection مرعبة للأهل و الدكتور بمجرد التفكير في تشخيصها

Meningococcal infection is a disease that strikes fear into both parents and doctors, as it can kill previously healthy children within hours.
Fortunately, it has become uncommon (due to introduction of the meningococcal vaccine) Can cause meningitis (meningococcal meningitis), or rapidly progressive sepsis (meningococcemia).

- **Clinical picture:**
  Meningococcal septicemia: High grade fever 40, irritability, very toxic & ill – Petechial (purpuric) rash → rapidly progressive to S/S of septic shock (poor perfusion–delayed CRT –tachycardia–hypotension–confusion–coma)



- **Immediate treatment:**
  - ABC – Call for help.
  - Immediate empirical antibiotics 3rd generation cephalosporin (ceftriaxone IV)
  - ICU management
  - Isolation, notification, prophylaxis for contacts.

1. Any febrile child with a purpuric rash or who is very unwell should be given intramuscular benzylpenicillin or intravenous third-generation cephalosporin before urgent transfer to hospital.
2. Meningococcal septicaemia can kill children in hours. Optimal outcome requires immediate recognition, prompt resuscitation and antibiotics.
3. meningococcal infection has the lowest risk of long-term neurological sequelae, with most survivors recovering fully, though limb loss sometimes occurs.

Encephalitis

- **Definition:** Is an inflammation of the brain parenchyma.
- **Organisms cause encephalitis by:**
  1. Direct invasion of brain by neurotoxic virus (such as HSV)
  2. an apparent immune mediated response in CNS begins several days after extra neural infection (postinfectious encephalopathy), e.g. following chickenpox.

- **Etiology:**
  - **Viruses most common: Enteroviruses*****, respiratory viruses (influenza viruses), arboviruses and herpesviruses e.g. **HSV**, varicella zoster virus (VZV), and human herpesvirus 6 (HHV–6)

Herpes simplex virus (HSV) is a very rare cause of childhood encephalitis but it may have devastated long term consequences. (MR = 70% in untreated patients).
All children with encephalitis should therefore be treated initially with high–dose intravenous acyclovir until this diagnosis has been ruled out, because this is a very safe treatment.

  - **Some bacterial causes:** Mycoplasma, B. burgdorferi (Lyme disease), Bartonella henselae (cat scratch disease), rickettsial infections (e.g. Rocky Moun– tain spotted fever).

- **Clinical picture:** **insidious** onset
  - **Non–specific symptoms:** (Fever– Cough– Vomiting– Headache– Rash)
  - **Neurological symptoms:** S/S of increased ICP – Lethargy – Behavioral Changes –Convulsions– Neurological deficits.
- **Diagnosis:**
  - CBC (May be Normal), CRP
  - CSF Examination: (Clear appearance – Lymphocytosis – N or high protein – N sugar)
  - EEG.
  - CT scan or MRI Brain (Show focal changes, particularly within the temporal lobes (HSV encephalitis).
  - Serology & PCR ↑ Specific & Diagnostic.

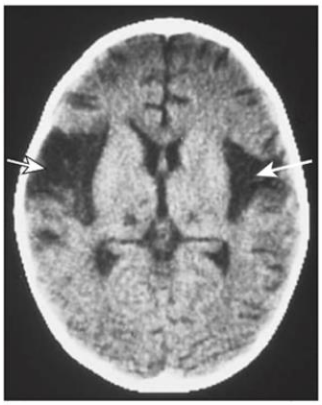


Figure 15.5 Herpes simplex encephalitis. The computed tomography scan shows gross atrophy from loss of neural tissue in the temporoparietal regions (arrows).

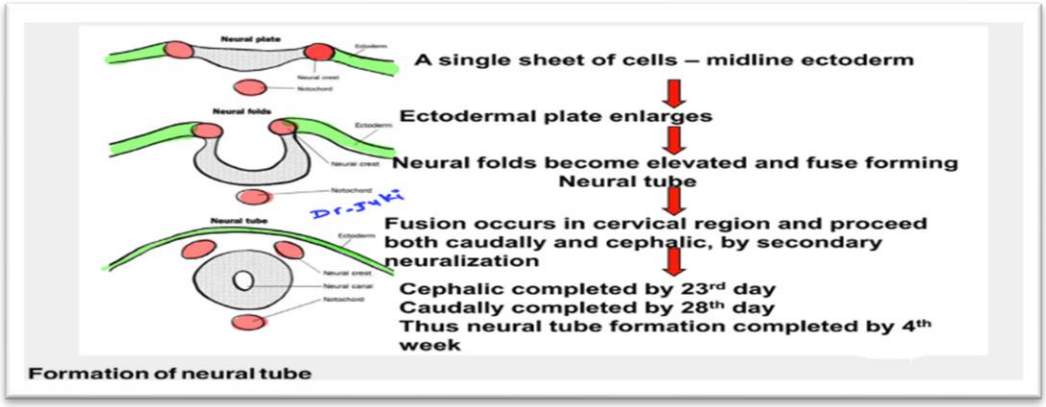
- **Treatment:**
  - Supportive Rx (Admission to PICU, Control convulsion (Anticonvulsant), IVF (2/3 m), Monitoring
  - Specific Rx: Aciclovir IV (DOC) For HSV encephalitis for 3 weeks

- In encephalopathy from a noninfectious cause, such as a metabolic abnormality, the clinical features may be similar to infectious encephalitis.
- The underlying causative organism is only detected in <50% of the cases in the UK.

Indication of 2/3 Maintenance IVF (IVF restriction)
- Meningitis
- Encephalitis.
- Congenital Heart disease
- Birth asphyxia
- Head trauma.
- Severe pneumonia.

Neural tube defects

Neural tube defects result from failure of normal fusion of the Neural plate to form the neural tube during the 1st 28 days following conception.



- **Causes:** (Multifactorial) Genetic + Environmental Factors
- **Risk Factors:**

1. Folic acid deficiency
2. Maternal DM.
3. Maternal anticonvulsant use.
4. Maternal smoking –obesity

   ✓ The risk of having a second affected infant is 3–5%
   ✓ Mothers of a fetus with a neural tube defect have a 10–fold increase in risk of having a second affected fetus.

- **Types:**
  1. **Anencephaly:** Is failure of development of most of the cranium & brain – either stillborn or die within a few hours – detected on antenatal ultrasound screening.
  2. **Encephalocele:** Herniation of sac, which may contain the brain through a midline skull defect, (most are occipital), usually associated with brain malformations.


Anencephaly Encephalocele

3. Spina bifida:

A. Spina bifida occulta. Defect in the vertebral arch due to failure of union of the Lamina (No protrusion of meninges) with intact spinal cord

- **Clinical picture:** Asymptomatic, often incidental finding on X Ray.

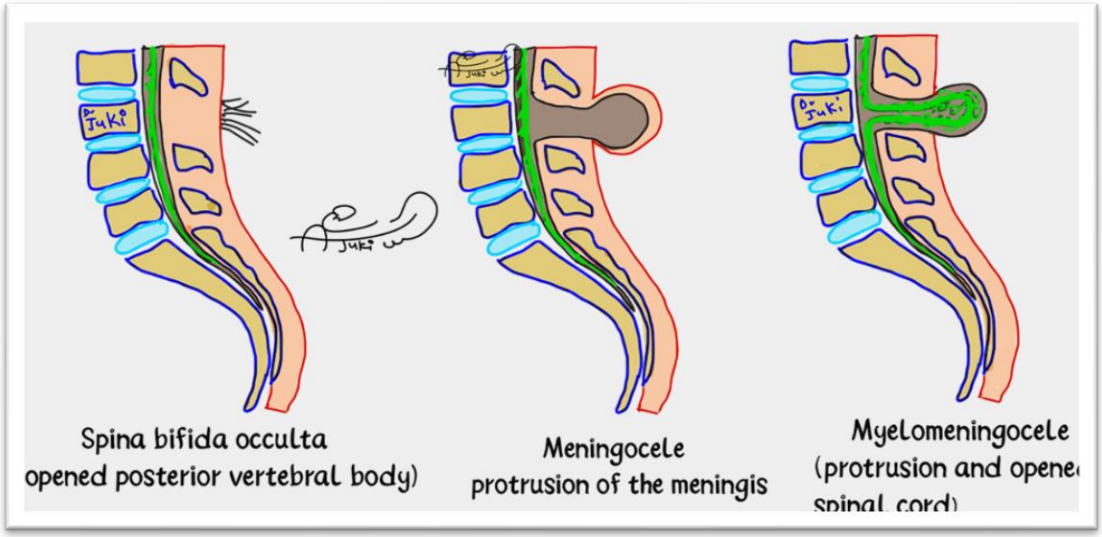
  It may be associated with overlying skin lesions such as tuft of Hair – Lipoma – Nevus, dermal sinus, hemangioma.



- **Site:** Usually in the lumbar region (S5–L1) & no neurologic deficit at birth but tethering of the spinal cord (diastematomyelia) may occur during childhood.
- **Investigation:** X ray or MRI Spin.

B. Meningocele:

In which the meninges protrude through a defect in the vertebral arch causing cystic swelling under skin, normal cord, usually have good prognosis after surgical repair.



C. Meningomyelocele:

In which the meninges & Spinal cord protrude through a defect in the vertebral arch.
The most common site is (Lumbosacral)


Meningocele


Meningomyelocele

- **Complication or association of meningomyelocele:**

    1. Deformities (DDH –Club foot – scoliosis)
    2. Paralysis of the lower limbs (LMNL) & hypotonia.
    3. Neuropathic bladder (bladder denervation) –palpable bladder, dribbling of urine predisposes to recurrent UTI, VUR & CKD.
    4. Neuropathic bowel (bowel denervation): patulous anus.
    5. Hydrocephalus from the associated **Arnold Chiari type 2 malformation** (herniation of the cerebellar tonsils and brainstem tissue through the foramen magnum), leading to disruption of CS flow, 80% of affected infants require a ventriculoperitoneal shunt.
    6. Sensory loss.

- **Treatment:**
  The lesion was repaired soon after birth & needed a **multi–disciplinary team**.
  Neurosurgery, Physiotherapy, Urology, Nephrology, Orthopedics.

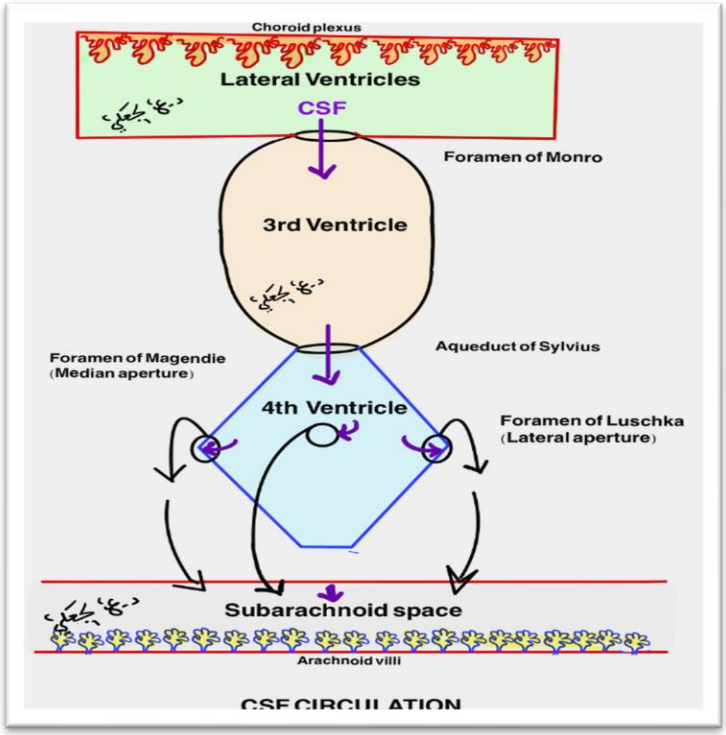
- **Prevention:**
    - All females: Folic acid 0.4 mg daily.
    - High–risk females (previously affected child): Folic acid 4 mg daily.

  Supplementation should be started 1 month before plan conception

Hydrocephalus

- **Definition:** Enlargement Of cerebral ventricles due to excessive accumulation of CSF. (CSF volume — 50 ml in neonate – 150 ml in Adult)

- **CSF Circulation:**



- **Mechanism of ventricular dilatation:**

1. Increase CSF production
2. Defect in absorption
3. Obstruction anywhere in the ventricular system.

Obstructive (non-communicating)	Non-obstructive (Communicating)
1_Congenital Malformation: 1. Aqueduct of Sylvius stenosis. 2. Dandy – walker malformation: It's due to congenital 3. Atresia of foramina of Luschka & mogendi Lead to dilatation of the 4th ventricle (Prominent occiput) 4. Arnold – Chiari Malformation: (Type II): Congenital downward displacement of the cerebellar tonsils & medulla into the foramen magnum often associated with spina bifida (Meningomyelocele). 5. Toxoplasmosis (Congenital infection) **2_Aquired causes:** 1. Post meningitis. 2. Post hemorrhagic (Intraventricular hemorrhage in preterm infant) 3. Posterior fossa neoplasm, vascular malformation	**1-CSF Overproduction** (Choroid plexus papilloma) **2-Defective reabsorption:** 1. Subarachnoid hemorrhage 2. Meningitis, e.g. pneumococcal, tuberculous 3. Dural sinus thrombosis.

- **Clinical features:**



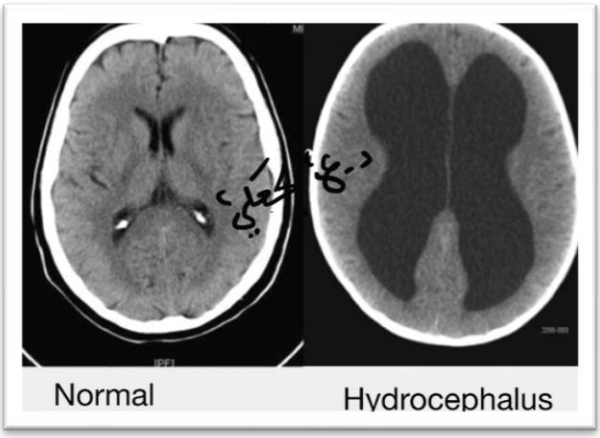
1_In Infant:
- Large head (Disproportional to the body)
- Fontanels are widely open & bulging
- Sutures widely separated.
- Visible dilated veins.
- An advanced sign is fixed downward gaze or "**sun setting**" of the eyes

2_In old child:-

Symptoms of increased ICP (Frontal headache– Projectile Vomiting–photophobia)

Signs
- Large head – Dilated veins – sunset eyes.
- Large & delayed closure fontanelle – wide sutures.
- Percussion: Macewen sign.
- Craniotabes (+ve. Ping pong sensation)
- Bruit on Auscultation of AF (Av malformation) Transillumination test (+ve)
- Any infant with hydrocephalus (Don't forget to examine Back for any NDT & motor examination of the lower limbs

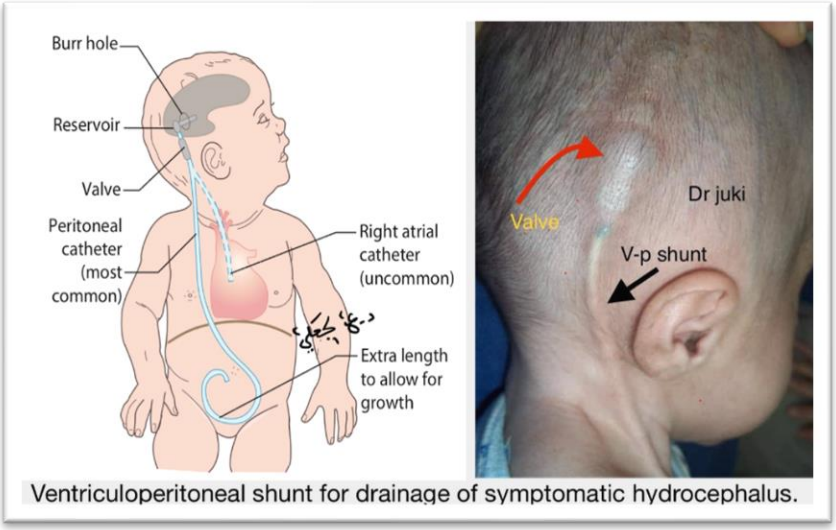
- **Diagnosis:**
  - Antenatal: Congenital type can be diagnosed antenatal by Ultrasound screening.
  - Neonate & Infant:
    1. U/S head (Ventricular dilatation)
    2. CT or MRI brain scan (+ degree of cortical Atrophy + may delete the cause)
    3. OFC (Head circumference) Should be monitored overtime on the centile chart.
    4. Karyotyping if chromosomal anomalies suspected
    5. TORCH screen.



- **Treatment:** The mainstay is the surgical insertion of a **ventriculoperitoneal shunt (VP Shunt).**

- **Complications of VP shunt:**

  1. Infection (Staph epidermidis)
  2. Malfunction due to blockage.
  3. Shunt nephritis (Immune complex mediated nephritis after prolonged infection with staph epidermidis
     - C/P:
       Hematuria, proteinuria – high BP –Low complement.

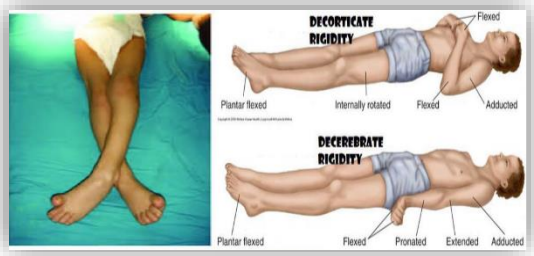
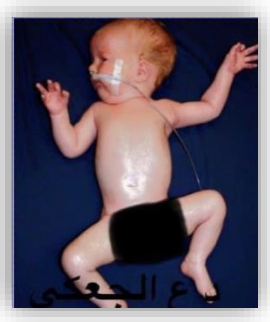
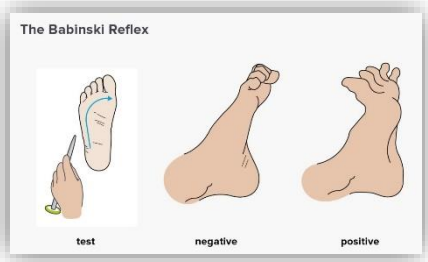


Ventriculoperitoneal shunt for drainage of symptomatic hydrocephalus.

- Other options of treatment (Endoscopic 3rd ventriculostomy – External ventricular drainage)

- **Arrested hydrocephalus**. (Non progressive hydrocephalus –Usually occur post meningitis– Needs regular follow up).

Motor Neuron Lesion

مهم جدا كلينك	Upper motor neuron lesion	Lower motor neuron lesion
Definition	Any lesion above the anterior horn cell in the spinal cord or above the nuclei of the cranial nerves (Cortex –cerebellum –basal ganglia –pyramidal or extra pyramidal tract).	Any lesion in anterior horn cell, or below it in motor nerve fiber or neuromuscular junction
Inspection Posture	 scissoring –decorticate –decerebrate posture	hypotonic posture (frog like) Positive Fasciculation 
Muscle wasting (Atrophy) Abnormal movement	No / + due to disuse Seizures or chorea –athetosis	
Power	paralysis or weakness **Grades of Ms power مهم كلينك** 0. No visible contraction, no movement 1. Visible muscle contraction, but no limb movement 2. Active movement, but not against gravity 3. Active movement against gravity 4. Active movement against gravity and partial resistance 5. Active movement against full resistance	paralysis or weakness.
Tone	hypertonia (spasticity or rigidity)	hypotonia
Reflexes (deep)	exaggerated (hyperreflexia)	hypo reflexes or areflexia
Babinski sign (superficial plantar response) Ankle clonus	positive  positive	negative negative
example	1. cerebral palsy 2. Stoke, ICH. 3. Neurodegenerative disease 4. Brain tumors	1. Spinal Ms atrophy 2. Guillain–Barre syndrome 3. Myasthenia gravis. 4. Myopathy, DMD.

Cerebral palsy

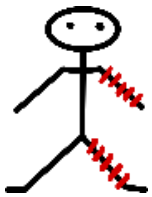
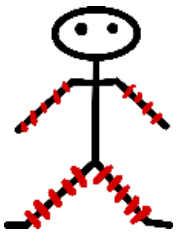
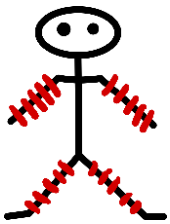
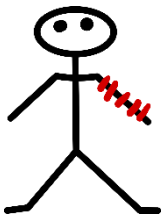
- **Definition:** is disorder of **movement and posture** caused by **nonprogressive** insult or injury to the **developing** brain. Non-fatal, non-curable, non-progressive).
  - The CP is the **most common** cause of neurodisability in childhood
  - **prevalence** of 1-2 / 1000 live births.

- **Cause:**

Antenatal 80%	Natal 10%	Postnatal 10%
• Congenital infection, • Congenital brain malformations • Radiation, Toxins (Alcohol, cocaine) • Cord prolapse, abruptio placenta	• Hypoxic ischemic encephalopathy	• Early meningitis, encephalopathy • Trauma (LCH), kernicterus • Severe hypoglycemia, hydrocephalus

- **Clinical Presentation:**
  - Abnormal limb tone and/or trunk posture, delayed motor milestones.
  - Feeding difficulties (slow feeding – gagging – Vomiting – aspiration)
  - Abnormal gait.
  - Asymmetric hand function < 12 months age.
  - Persistence of moro & asymmetric tonic neck reflex > 6m.
  - Early hand preference (Handedness) < 12 m.
- **Types:**

  1-Spastic cerebral palsy: 80-90% (Commonest type)
  - Damage to pyramidal tract or corticospinal tract (UMNL)
  - Tone: hypertonic (Spasticity, clasp knife' fashion.
  - Power — paralysis.
  - Reflex — Hyperreflexia + Babinski sign (+ve)

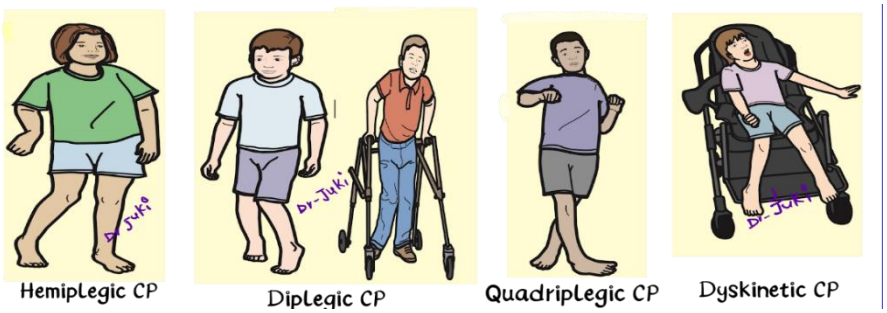
Hemiplegic CP	Diplegic CP	Quadriplegic CP	Monoplegic CP
unilateral involvement of the arm and leg, with the face spared.  often present at 4-12 months of age with fisting of the affected hand, a flexed arm, a pronated forearm, asymmetric reaching, hand function, later affected side tiptoe walk (toe-heel gait) Some cases caused by neonatal stroke	all four limbs, but legs are affected to a much greater degree than the arms, Walking is abnormal.  Usually associated with preterm birth due to **periventricular** brain damage	all four limbs affected severely, the trunk is involved with a tendency to opisthotonus, poor head control and low central tone  a more **severe** form of CP is often associated with seizures, microcephaly and moderate or severe intellectual impairment, MR. H/O perinatal hypoxic-ischemic encephalopathy	a rare type of cerebral palsy, affecting only one limb either upper or lower 

2–Dyskinetic cerebral palsy: 10% (choreoathetoid CP)

- Damage to the extrapyramidal tract. (Basal ganglia)
- Tone — hypertonia (rigidity)
- Power — Paralysis
- Reflex — hyperreflexia
- associated with involuntary movement as **chorea** — **athetosis** (is slow writhing movements), **dystonia** (is simultaneous contraction of agonist and antagonist muscles), Intellect may be relatively unimpaired.
- often present with floppiness, poor trunk control and delayed motor development in infancy.
- seen in patients with (Kernicterus– hypoxic–ischaemic encephalopathy).

3–Ataxic cerebral palsy:

- Damage to the cerebellum.
- Incoordination of movement _ Intention tremor _ ataxic gait.
- There is early trunk and limb hypotonia, poor balance and delayed motor development. Incoordinate movements, intention tremor and an ataxic gait may be evident later.



4–Atonic cerebral palsy.

5– Mixed C.P.

- Diagnosis: (mainly clinically).

- Investigation: To determine etiology & exclude other causes.

1. Brain imaging: Cranial US. CT scan. MRI Brain to identify Malformation & neurodegenerative brain disease.
2. TORCH screen.
3. Metabolic screen.
4. EEG if seizure

- Common problem (complication) associated with C.P:

CNS	RESPIRATORY	GIT	MSK SYSTEM	SKIN	SOCIAL
1. MR & intellectual disability 60–70% 2. Epilepsy 30–40%. 3. Disability & developmental delay 4. Visual problems. 5. Speech & hearing problems.	1. Aspiration pneumonia 2. Recurrent chest infection	1. Feeding difficulties. 2. FTT 3. GERD 4.Constipation	1. Ms pain & spasm, joint subluxation. 2. Ankylosis of joints, stiffness. 3. joint contracture & deformities	1. Bed sores & skin infection	1. Social problems in the family. 2. Behavior problems

- **Treatment:** Needs a **multidisciplinary team (MDT)**
  - Pediatrician.
  - Physiotherapist
  - Neurologist: Anticonvulsant if there is convulsion
  - Speech & Language therapy.
  - Orthopedic: Ms spasticity treated by: Ms relaxant (Baclofen), botulinum toxin injection, or deep brain stimulation of the basal ganglia, joint contractures may need surgical correction.
  - Nutritional therapy: care for feeding.
  - social worker & Rehabilitation.

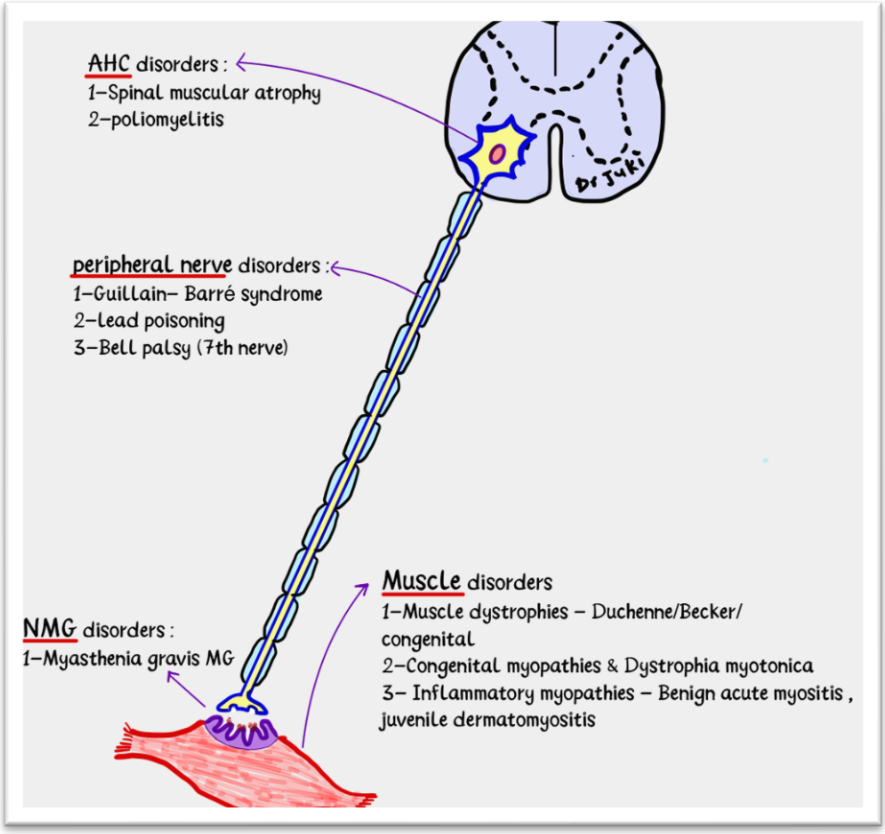
Parents should be given details of the diagnosis as early as possible, but prognosis is difficult during infancy until the severity and pattern of evolving signs and the child's developmental progress have become clearer over several months or years of life

	Cerebral palsy	Neurodegenerative disease
• Course • Family history • Head size • Hepatosplenomegaly • Skin sign	• Non progressive • – ve • Normal or small • No • No or bed sores	• Progressive • +ve • Larg (Usually) • Yes. • Yes, as Cafe au lait spots. or Ash leaf spots.

Neuromuscular disorders

Any part of the lower motor neuron pathway can be affected in a neuromuscular disorder, (**anterior horn cell disorders, peripheral neuropathies, disorders of neuromuscular transmission and primary muscle diseases**)

The key clinical feature of a neuromuscular disorder **is weakness, which may be progressive or static.**



Spinal muscular atrophy (SMA)

It is degenerative disease of AHC in the spinal cord & brain stem (motor cranial nuclei)

- Types:

Type	synonyms	Onset & severity
Type 1 SMA	Werdnig Hoffmann disease (Acute infantile)	from birth to 6 months (severe)
Type 2 SMA	Late infantile	between 6 months to 18 months (intermediate)
Type 3 SMA	juvenile =Kugelberg-Welander	after 18 months (mild)
Type 4 SMA	adult onset	mid 30 (mild)

The most severe form is SMA type 0, diagnosed in newborn infants, survival rate is limited

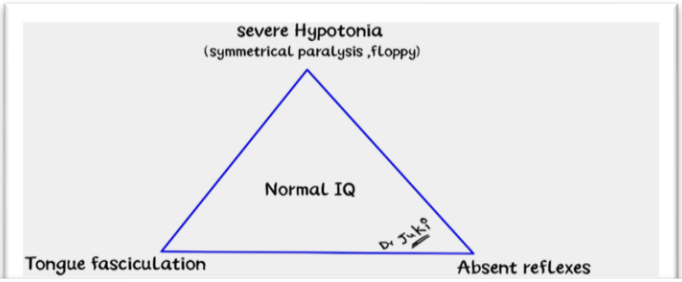
Werdnig Hoffman disease

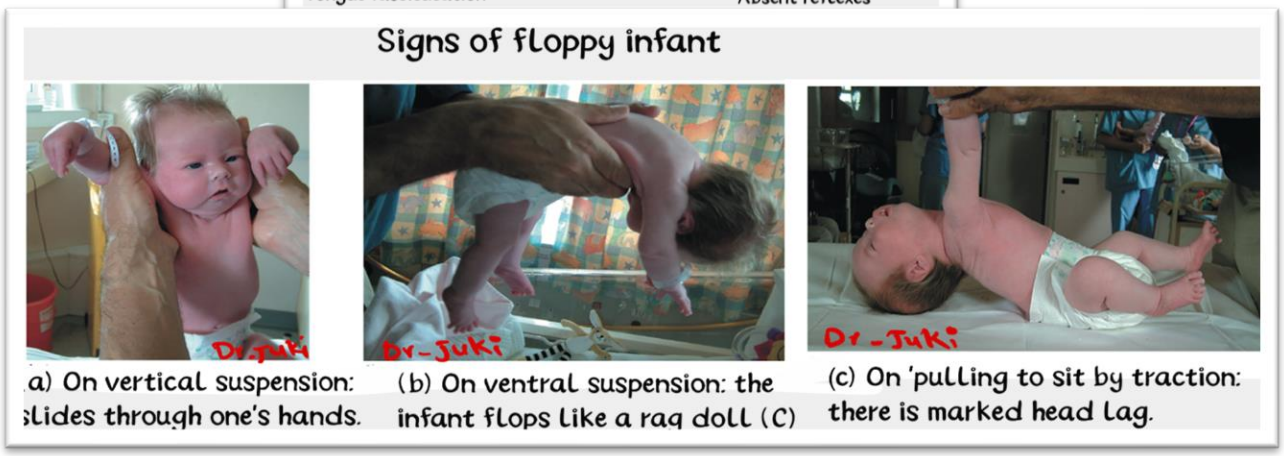
- Autosomal recessive disorder (**AR**)
- The **2nd most** common cause of neuromuscular disease in the UK after Duchenne muscular dystrophy.
- It's the most common cause of floppy infants.

- Incidence: 1:10,000 Live births.

- Pathogenesis:
  It is caused by **mutations** of the survival motor neuron gene **SMN1** on chromosome 5, that code for SMN Protein → Diminished abundance of this protein leads to death of neuronal cells in the **AHC** of the spinal cord (Progressive destruction of AHC).

- Clinical Pictures:
  - Antenatal: **decreased fetal movement** in 30% of cases.
  - Postnatal: A very severe progressive disorder presenting from birth to 3 months of age





- **Bulbar** palsy: weakness of bulbar muscles causing weak cry and poor suck with pooling of secretions.
- **Respiratory** muscle involvement: weak respiratory Ms & weak cough → accumulation of secretions →**recurrent** chest **infections**
- Also, weak respiratory Ms (see saw respiration) → recession, RD, cyanosis & atelectasis
- Finally **respiratory failure** & death.
- These children never sit unaided. Death is from respiratory failure within about 12 –18 months of age.
- **Facial & extra ocular** Ms are spared → alert expressions.
- **Arthrogryposis**: (positional deformities of the limbs due to antenatal compression may present)

- Investigation:

  - Creatine kinase — Normal level
  - EMG — shows Fasciculation & Denervation

- Muscle biopsy — to differentiate it from other neuromuscular disorders
- **Genetic testing** — can be done prenatal & postnatal.***

- **Treatment:**

  1–supportive Rx:
  - Respiratory care: (Ovoid aspiration – physiotherapy–ttt infection –respiratory support ,MV)
  - Nutrition: NGT or gastrostomy feeding high calories.
  - Rx orthopedic complication
  - Genetic counselling, social support.

  2–Specific Rx:
  - **Gene therapy** is now available in some countries but is very expensive.
  - Nusinersen SPINRAZA (partially functioning gene, SMN2)
  - Onasemnogene ZOLGENSMA

Guillain–Barre syndrome

Acute post – infectious polyneuropathy.

- **Definition:** Is an acute post infectious autoimmune demyelinating polyneuropathy. (mainly affects motor nerves but sensory & autonomic nerves sometimes affected)

- **Etiology:** Autoimmune demyelinating polyneuropathy usually 2–3 wks after URTI or gastroenteritis:
  - **Viral** (EBV – CMV – HIV)
  - **Bacterial** (Campylobacter – mycoplasma pneumonia).

- **Age:** 3–12 years (Can affects all age groups)

- **Clinical picture:** H/O preceding infection.

  1–Motor: Ascending flaccid paralysis + areflexia.
  - Bilateral & symmetrical usually.
  - Ascending: Lower limbs → trunk → Upper limbs…..
  - Progression of symptoms develops over hours to days.
  - Bulbar ms may affected lead to difficulty in chewing, swallowing & Risk of aspiration
  - Respiratory ms paralysis lead to respiratory depression & respiratory failure
  - Cranial nerves involment: bilateral facial weakness is easily missed in young children.

  2–Sensory: distal paresthesia & tingling in the toes & fingers (may be the earliest symptoms).

  3–Autonomic: **Dysautonomia** occurs in 70% and manifests as HR (tachycardia, bradycardia and other arrhythmias), BP (hypertension and orthostatic hypotension), urinary retention, ileus, and loss of sweating.

- **Investigation:** (Diagnosis mainly clinically)

  1. Lumbar puncture: CSF shows (Cyto albuminous dissociation: High CSF protein but normal cells). (In 2nd week).
  2–EMG: Shows demyelination of the peripheral nerves.
  3–Nerve conduction studies: nerve conduction velocity is reduced (In 2nd week).
  4–Stool analysis, C&S — to exclude other causes. (Poliomyelitis)
  5–MRI spinal cord (or brain and spinal cord) is the most useful acute investigation, to identify or exclude a spinal cord lesion, e.g. a bleed, tumor, or transverse myelitis.

- **Treatment:**

SUPPORTIVE TREATMENT	SPECIFIC TREATMENT
• Hospital admission. • Respiratory support & monitoring. • ICU care in severe cases (nutrition – IVF – monitoring –bladder care) • Physiotherapy	• Intravenous immunoglobulin (DOC) Dose (1–2g/kg) for 1 or 2 days slowly • Plasmapheresis (plasma exchange) NB (Corticosteroids have no beneficial effect and may delay recovery)

- **Prognosis:**
  - Complete recovery in ≃ 85–90% starting within 2–3 weeks (in a descending manner), although full recovery may take up to 2 years.
  - Residual weakness in some patients
  - Mortality <5% mainly due to respiratory failure.

Miller–Fisher syndrome: Acute external ophthalmoplegia, Ataxia & Areflexia.
Chronic variant: Chronic inflammatory demyelinating polyradiculopathy (CIDP) also called (relapsing or unremitting)

❖ GBS Vs Poliomyelitis.

GBS	Poliomyelitis
• Symmetrical paralysis. • Paralysis usually over 1–2 wks • Motor, sensory, autonomic nerves May be affected • Pathology: peripheral Nerve (Demyelination) • CSF findings: normal cells –high protein. • No virus isolated. • Residual paralysis: No in most cases. • Common cause of AFP.	• Asymmetrical paralysis. • Paralysis is usually over 1–2 days. • pure motor (No sensory, no autonomic involvement) • Pathology: A.H.C destruction • CSF findings: high Cells – high/ N protein. • Virus can be isolated (polio virus) • Residual paralysis: yes • Rare (effective vaccination).

The muscular dystrophies

This is a group of inherited disorders with progressive muscle degeneration (Genetic –progressive degeneration –primary)

Duchenne muscular dystrophy

- The **most** common muscular dystrophy (1:3000–6000) boy.
- **X Linked recessive** (male affected & Female carrier) (≃ 30% de novo mutations)
- It results from a deletion of the **gene for dystrophin,** (Locus: Xp21) which connects the cytoskeleton of a muscle fiber to the surrounding extracellular matrix through the cell membrane. Where it is deficient, there is gradual, progressive myofiber necrosis. (**Progressive proximal muscle weakness**)

- **Clinical picture:**
  - At about 2 to 3 years the boy develops unstable walking & inability to run properly & frequent falls.
  - Waddling gait at 5 years.
  - Gower sign (The child typically arises from lying position) on the floor by using his arms to climb up his legs & body.
  - Pseudohypertrophy: Calf muscle (2ry to replacement of degenerated muscle by fate & fibrous tissue)



Calf Ms Pseudohypertrophy



Gower sign positive

  - Wheelchair by 12 years of age.
  - Mentality is normal or slightly impaired.
  - Association: * **Cardiomyopathy (HF)– * Scoliosis is a common – * About ⅓ of cases have Learning difficulties & language delay.**

- **Investigation:**
  1. Serum creatine phosphokinase (CPK): Markedly elevated.
  2. Muscle biopsy: shows muscle fiber degeneration & replaced by Connective tissue.
  3. Genetic testing (DNA): Diagnostic (prenatal Dx possible)
  4. EMG: shows myopathic changes.
  5. CXR, ECG, Echocardiography (cardiomyopathy)
  6. Some countries (but not the UK) have neonatal screening programmes/ affected children are detected by an elevated plasma creatine kinase (CK).

- **Treatment:**
  - Rx mainly supportive (Multidisciplinary)
  - Physiotherapy & splinting of the ankles helps to prevent contractures, Achilles tendon lengthening and scoliosis surgery may be required
  - Respiratory support: Overnight CPAP (continuous positive airway pressure)

- Medical Rx: steroids may help to preserve mobility and prevent scoliosis.
- Genetic counselling & molecular genetic therapies are becoming available nowadays.

Prognosis: DMD is **progressive** disease, Life expectancy is reduced to the late twenties (**25 years**) from **respiratory failure or the associated cardiomyopathy.**

Backer's muscular dystrophy:

- It's similar to DMD but clinically the disease is **milder** and progresses more **slowly.**
- Low dystrophin (some functional dystrophin present)
- Onset by the age of 11 years, inability to walk at twenties & death at 40 years old.

Dystrophia myotonica type 1:

The 2nd most common muscular dystrophy.

- **Etiology:** AD (It is caused by Triplet Repeat Expansion of (CTG)

  A) Neonatal form:
  - Hypotonia, Feeding & respiratory complications.
  - Decreased Fetal movements & oligohydramnios antenatal.

  B) infantile & late childhood:
  - myopathic facial appearance: Concave temporalis, Inverted V-shaped upper lip & high-arched palate
  - Skeletal muscles: Weakness??
  - **Myotonia** (=Slow relaxation after muscle contraction) develops after the age of 3 yrs
  - Cardiac: Heart block & arrhythmias (cause of death)
  - Cataract & frontal baldness (adults)
  - Intellectual impairment
  - Endocrinal: DM, delayed puberty, Adrenocortical insufficiency, Hypothyroidism



Myotonic dystrophy : marked facial weakness and moderately severe learning difficulties.

- **Investigation:**
  - Serum CK: Normal (or mildly increased)
  - NCV: Normal
  - EMG: myotonic pattern
  - ECG
  - Genetic analysis & antenatal diagnosis
- **Treatment:** (Supportive)

Myasthenia gravis

- **Definition:** Is an autoimmune disease which antibodies to acetylcholine receptors at the neuromuscular Junction block & damages the neuromuscular junction.

Juvenile myasthenia	Transient neonatal myasthenia	Congenital myasthenic syndromes
Due to: Anti-ACh receptors antibodies **Clinical picture:** (>10 years) • Easy fatigability on repetition of movements, • Ocular muscles: ptosis (usually asymmetric), diplopia. • Facial weakness (loss of facial expression, and difficulty chewing) • Bulbar symptoms: dysphagie • Skeletal muscles (Proximal > Distal) • Diurnal variation (worst at the end of the day) • Note (Sensation & Reflexes are Normal) **Diagnosis:** 1. Edrophonium test (Tensilon test) IV edrophonium: improves strength & Fatigability within minutes. 2. Antiacetylcholine receptor antibodies + ve in 60%–80% or, more rarely, anti-MuSK (anti muscle-specific kinase) antibodies will confirm the diagnosis  Myasthenia gravis showing ptosis from ocular muscle fatigue which improved with edrophonium. **Treatment:** • Acetylcholinesterase inhibitors (Pyridostigmine or neostigmine) • Plasmapheresis (in crisis) • Thymectomy is indicated if a thymoma is present. • Prednisolone & immunosuppressive therapy.	**Due to:** Transplacental passage of Ab (Myasthenic mother 10–20 %). **Clinical picture:** • ptosis, ophthalmoplegia • bulbar (feeding difficulties) • respiratory muscle weakness • Symptoms persist for 1 to 10 wks (Mean 3 wks) • All neonates have + ve Ab. **Treatment:** Transient forms need Anti-Cholinesterase for a few weeks.	it's rare genetic syndromes (No Antibodies) **Clinical picture:** • ptosis, ophthalmoplegia • bulbar (feeding difficulties) • respiratory muscle weakness. • arthrogryposis. • Non immune mediated. (No Ab). **Treatment:** do not always respond to anticholinesterase inhibitors.

Juvenile Dermatomyositis: ((inflammatory myopathies))

systemic illness, probably due to an **angiopathy**.

Age of onset: 5–10 years.

Clinical picture:

- Fever
- muscle: symmetrical ms weakness & pain (mainly proximal) +– dysphagia
- Skin Rash:
  1–Heliotrope rash violaceous (pink–purple) rash on the eyelids + periorbital edema,
  2–Subcutaneous calcification on the extensor surface (calcinosis).
  3–Gottron papules over the metacarpophalangeal joints.



Heliotrope rash



Calcinosis cutis

Investigations:

- CRP – ESR (sometimes raised)
- CPK (high)
- Muscle biopsy: Inflammatory cell infiltrate + atrophy

Treatment:

- Physiotherapy
- Steroids (DOC) For 2 years.
- Immunosuppressants drugs eg. methotrexate, cyclosporine.

Prognosis: MR 5 – 10%

The (Floppy infant)

The floppy infant describes an infant with marked hypotonia.

Clinically:

- Frog position.
- Marked head lag. (on traction of the arms)
- Curved trunk (c shaped) (rag doll) on ventral suspension
- Slide through one's hands when held upright (vertical suspension)
- Hypotonia on examination of tone.

Signs of floppy infant



(a)Frog like posture



(b) On vertical suspension: slides through one's hands.



(C) On ventral suspension: the infant flops like a rag doll (C)



(d) On 'pulling to sit by traction: there is marked head lag.

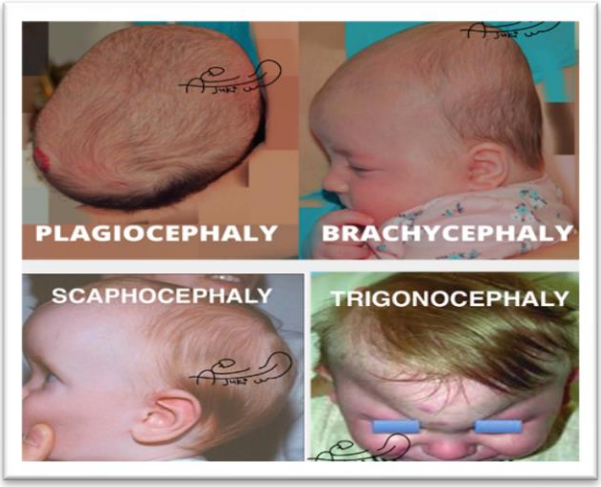
Causes of hypotonia :–

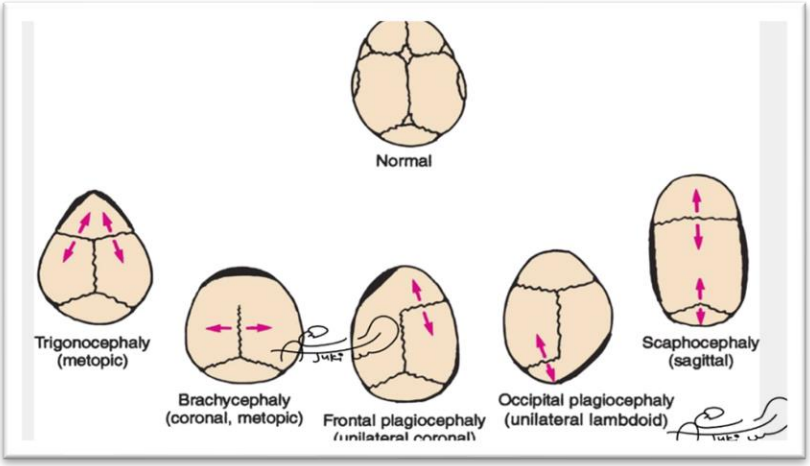
1. Central hypotonia	2. Peripheral hypotonia
1– CNS: •Cerebral malformation •Ischaemic/haemorrhagic brain lesions: (Hypoxic–ischemic encephalopathy HIE, Intracranial hemorrhage). **2–Metabolic:** Inborn errors of metabolism **3–Endocrine disorders** as congenital hypothyroidism **4– Chromosomal/** syndromes as Trisomy 21 & Prader–Willi syndrome	**1–Anterior horn cell:** •Spinal muscular atrophy type 1 (Werdnig–Hoffmann syndrome) **2–Neuromuscular junction:** • Neonatal myasthenia gravis **3–Muscles:** • Congenital myopathies. • Myotonic dystrophy.

Head Abnormalities

A. Abnormal head shapes:

1. **Brachycephaly**: Flat occiput e.g Down syndrome
2. **Scaphocephaly** (Dolichocephaly): Long, narrow head.
3. **Plagiocephaly**: asymmetrical head (Flat one side), usually is positional
4. **Acrocephaly** (Oxycephaly): elongated head upward.
5. **Trigonocephaly**: Keel – shaped forehead, Hypertelorism.





Craniosynostosis: Premature fusion of one or more sutures (craniosynostosis) may lead to distortion of the head shape.

The sutures of the skull bones normally start to fuse during infancy but do not finally fuse until late childhood.

Types:

1- Localized:

- common, it affects one suture, results in an abnormal shape as mentioned above.

2-Generalized:

- Multiple sutures resulting in microcephaly and developmental delay
- Sometimes associated with syndromes, e.g. Apert syndromes, Crouzon syndrome (exophthalmos)

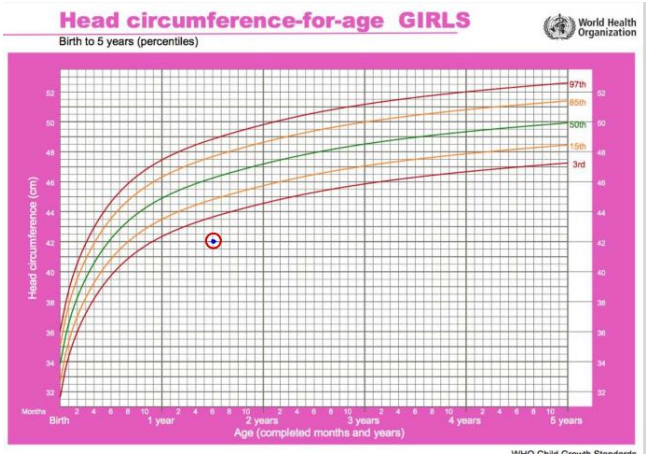
Investigations: skull X-ray or cranial CT SCAN

Treatment: surgically in specialist centers for craniofacial reconstructive surgery if there is raised intracranial pressure, or for cosmetic reasons.

Abnormal head size

Microcephaly (Small head)

- OFC or head circumference below 3rd centile (WHO) for age sex. (or below the 2nd centile)
- Head circumference more than 2 standard of deviation SD below the mean for age & sex.



Causes:

1–Primary

- Familial (when it is present from birth and development is usually normal)
- Chromosomal & syndromes as trisomy 21–18–13.

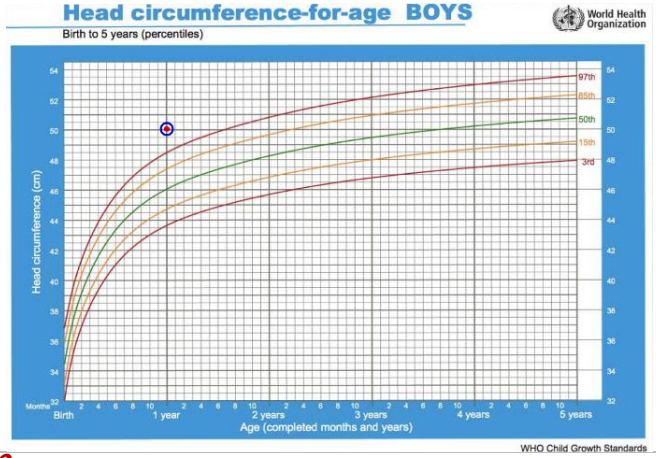
2–Secondary

- perinatal asphyxia, meningitis, or cerebral palsy,
- TORCH
- Irradiation exposure
- Metabolic (PKU)
- Maternal drug (fetal Alcohol syndrome)

3–craniosynostosis

Macrocephaly (Large head)

- OFC or head circumference above 97th centile (WHO)for age & sex (or above 98th centile)
- Head circumference more than 2 standard of deviation SD above the mean for age & sex.



Causes:

1. Familial
2. Hydrocephaly
3. Rickets –Thalassemia major.
4. Achondroplasia & Osteogenesis imperfecta.
5. metabolic disease as Mucopolysaccharidosis
6. Cerebral gigantism (Sotos syndrome)
7. High intracranial pressure (in an infant):
   a) chronic subdural hematoma
   b) brain tumor
   c) neurofibromatosis.

Brain Tumours

- Brain tumors are **almost** always **primary** in children.
- Brain tumors are 2nd most **common** tumor in pediatric (after Leukemia)
- Brain tumors are the most common **solid** tumor in children.

- Divided into: 1–**Infratentorial tumors 60% (Common)**
  2– Supratentorial tumor 40%

Clinical picture:

1_Sign of ↑ ICP:
- Headache (Persistent or recurrent)
- Persistent or recurrent Vomiting (projectile–Morning)
- Plugging anterior fontanelle – increase in head circumference– Developmental delay/regression (in infants)
- Eye sign: blurred vision– diplopia, squint, Nystagmus.
- Cranial nerve palsy (IV– IIV)
- Papilledema (late sign)

2_Focal neurological:
- Seizures
- Hemiparesis
- Ataxia (Problems with balance, coordination or walking)

3–Other features:
- Endocrinopathies – e.g. Delayed or arrested puberty, slow growth.
- Behavioral changes.
- Deteriorating school

- **Types:**

Astrocytoma 40%
- The most common brain tumor type.
- varies from (low grade) to highly malignant (high grade, glioblastoma multiforme).
- Usually cystic & slow growing.
- C/P: s/s of high ICP – hemiplegia –seizures–behavior changes.
- Diagnosis: MRI brain
- Treatment: Surgical resection, radiotherapy +/– chemotherapy.

Medulloblastoma 20%.
- Usually in midline (posterior fossa)
- C/P: s/s of high ICP + Ataxia–abnormal eye movements.
- 20% have spinal metastases at diagnosis.
- Treatment: surgical resection + Chemotherapy, Radiotherapy.

Brain stem glioma 6%
- C/P: ataxia – cranial nerve palsy, pyramidal tract lesion
- Diagnosis: MRI brain
- Treatment: mainly Palliative (Radiotherapy)
- Prognosis poor (<10–20% survival

Ependymoma 8%

- Locally invasive. Most common site is 4th ventricle
- ≃ 10% are slow growing low grade but the remainder present as aggressive tumors requiring complete resection and radiotherapy
- C/P: s/s high ICP + cranial nerve palsy.
- Treatment: surgery + radiotherapy.

Craniopharyngioma (4%)

- developmental tumor arising from the squamous remnant of Rathke pouch.
- It is not truly malignant but is locally invasive and grows slowly in the suprasellar region
- Solid of cystic areas (90% shows calcification on X Ray)
- C/P:
    - headache–s/s high ICP.
    - Visual field loss – bitemporal hemianopia
    - Pituitary failure – growth failure, diabetes insipidus, weight gain.
- Diagnosis: MRI brain
- Treatment: Surgery +/– Radiotherapy.
- Prognosis: good survival but risk of long–term visual impairment and lifelong, complex pituitary insufficiency.

Headache

Headache is a frequent reason for older children and adolescents to consult a doctor.

Types:

1–Primary headache

- **Tension headache:** a symmetrical headache of gradual onset, often described as tightness, a band or pressure. There are usually no other symptoms.
- **Cluster headache:** repeated attacks of very severe pain on one side of the head
- **Migraine:** is the most important & most frequent type of headache in children & adolescents (10%), +ve family history, triggers: Exercises, stress, school, menstruation, head trauma, dehydration, foods

  Pathogenesis: (Vasomotor instability).

TYPES of Migraine:

	1–Migraine without aura	2–Migraine with aura
incidence	common (90%)	rare (10%)
Aura	–ve	+ve can be: • Visual (Blurring, flashes of light…) • Sensory (paresthesia) • Dysphasic (Difficult verbal) • Motor (hemiplegia)
Clinical picture	• Unilateral or bilateral • pulsatile, over the temporal or frontal area. • Episodes lasts 1–72 hours. • accompanied by unpleasant GIT such as nausea, vomiting, abdominal pain or photophobia and phonophobia. • aggravated by physical activity and relieved by sleep.	• The headache is preceded by a unilateral aura (visual, sensory, or motor impairment), although the aura may occur without a headache • Visual aura: negative phenomena, such as hemianopia, scotoma or positive phenomena such seeing zigzag lines. • presence of premonitory symptoms (tiredness, difficulty

3–Uncommon forms of migraine: rare

- familial hemiplegic migraine – caused by a calcium • channel defect, dominantly inherited.
- sporadic hemiplegic migraine
- migraine with brainstem aura – vomiting with nystagmus and/or cerebellar signs
- periodic syndromes – often precursors of migraine (cyclical vomiting, abdominal migraine, benign paroxysmal vertigo of childhood).

Treatment:

- Analgesia paracetamol – NSAID.
- Antiemetic prochlorperazine or cyclizine for nausea
- Triptans (serotonin agonist) sumatriptan
- Physical treatment: cold compass – warm pads.
- Psychological support.

Prophylactic: if the headaches are frequent and intrusive:

- sodium channel blockers – topiramate or valproate
- beta–blockers – propranolol (contraindicated in asthma)
- tricyclics: pizotifen (5–HT2 antagonist) – can cause weight gain and sleepiness, dangerous arrhythmias in overdoes.

2–Secondary headache:

- Space occupying lesion (Brain tumor)
- Pseudotumor cerebri (idiopathic intracranial hypertension)
- Sinusitis – Eye problems.
- Infection – meningitis, encephalitis, abscess.
- Vascular disorder: HTN, high CO2, hyperpyrexia, convulsion.

Red flag symptoms & signs: (space occupying lesion)

Symptoms:

- Headache – worse lying down or with coughing or straining
- Headache – wakes up child or young person (different from headache on awakening, not uncommon in migraine)
- Associated confusion, and/or morning or persistent nausea or vomiting
- Recent change in personality, behavior or educational performance.

Signs:

- Growth failure
- Visual field defects – craniopharyngioma, Squint, Cranial nerve abnormality
- Fundi – papilledema
- Torticollis
- Abnormal coordination or gait – for cerebellar lesions.
- Bradycardia
- Cranial bruits – arteriovenous malformation

Bell palsy and facial nerve palsies

Bell palsy is an isolated **lower motor neuron paresis (LMNL)** of the **7th** cranial nerve leading to facial weakness

causes: often unclear.
but may be associated with herpes simplex virus (painful vesicles in the external ear) or Lyme disease.

Clinically:

Facial palsy of LMNL nature (Complete paralysis of the ipsilateral 1/2 of the face)

- Absence of forehead wrinkles
- Absent nasolabial fold
- Inability to raise the eyebrows
- Inability to close the eye (exposure keratitis)
- Deviation of the angle of the mouth to the healthy side
- Dropping of the angle of the mouth (with dribbling of saliva) – Inability to blow the checks
- Inability to show the teeth
- Loss of taste sensation of the anterior 2/3 of the tongue on the same side of the lesion



Treatment:

1. prednisone: if given during the first week, and speed full recovery.
2. Eye care: lubricating drops or ointment, a patch, or even tarsorrhaphy.
3. If herpes is suspected, treat with acyclovir.

Prognosis:

Recovery is complete in the majority of cases but may take several months.

There are important differential diagnoses.

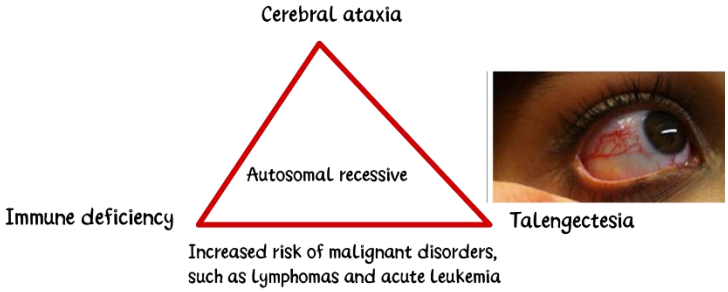
- Cerebellopontine angle tumor: if associated with 8th nerve paresis.
- brain stem lesion: if there is 6th nerve paresis, or ipsilateral cerebellar signs.
- coarctation of the aorta and renal failure: Hypertension should be excluded.

Ataxia

- Is Incoordination voluntary movement.
- **Types:** (1_Celebellar 2_Sensory 3_Vestibular)

- **Causes of cerebellar ataxia:** mcq

1. Inflammatory (Infection)
   - Cerebellitis as in cale of VZV – COX – Echovirus Usually Self-Limiting.
   - Bacterial ataxia (bacterial meningitis, cerebellar abscess).
   - Cerebellar Abscess –Encephalitis
2. Toxin or drugs: as Lead – Phenytoin – Alcohol –carbamazepine, lamotrigine
3. Traumatic
4. Brain Tumor – (Post fossa tumor)
5. Vascular (cerebral hemorrhage)
6. Migraine
7. Congenital Anomalies (Cerebellar agenesis/dysgenesis)
8. Ataxic cerebral palsy.
9. Metabolic – Abetalipoproteinemia – UIT E ↑
10. Hereditary: Ataxia Telangiectasia – Friedrich's Ataxia
11. Miller Fisher syndrome (a variant of Guillain–Barre syndrome).

Ataxia Telangiectasia mcq	Friedreich Ataxia
Genetic disorder of DNA repair **AR**, gene codes for a protein kinase mutation. **Clinical picture:**  	Rare **AR** Genetic disorder a triplet repeats in the FXN gene causing a lack of the frataxin protein **Clinical picture:** • Onset: age 10–12y • Gradual onset & Slowly prognosis cerebellar sign, ataxia and dysarthria, distal wasting in the lower limbs with absent reflexes and pes cavus. • Deep sensory loss & + Romberg sign. • Peripheral neuropathy • DM & cardiomegaly – heart failure – Death in 40–50yrs • No Effective treatment

Paediatric Stroke mcq

Brain stroke occurs in all age groups, may be due to (**vascular, thrombo– embolic or hemorrhagic disease**).

Etiology:

1. **CVS:** congenital cyanotic heart disease, e.g. Fallot tetralogy, endocarditis.
2. **BLOOD:** sickle cell disease/ deficiencies of antithrombotic factors, e.g. protein S.
3. **POSTINFECTIVE:** following varicella or other viral infections.
4. **INFLAMMATORY:** autoimmune disease, e.g. SLE (systemic lupus erythematosus).
5. **METABOLIC OR GENETIC:** homocystinuria, mitochondrial disorders.
6. **VASCULAR MALFORMATION** – arteriovenous malformation or moyamoya disease, in which there is a progressive involution of cerebral arteries. Moyamoya comes from the Japanese for 'puff of smoke', similar to the blurred appearance seen on angiography
7. **TRAUMA** – dissection of carotid or vertebral arteries

Investigations: MRI or CT scan brain– MR angiogram–CBC – coagulation profile –and vasculitis screen, and metabolic tests.

Treatment: **Rehabilitation** requires the involvement of the **multidisciplinary** therapy team. Low–dose **aspirin** prophylaxis is recommended after arterial ischemic stroke.

Autistic spectrum disorder

- Autistic disorder is **developmental** disorder
- Autistic disorder is characterized by marked **impairment in social interaction** & **communication** & **restricted range of activities** & **interests**.
- Impaired social interaction is the **hallmark** of this illness.

Cause: Unknown. Multifactorial (< 10% organic cause)

Epidemiology:

- **Incidence:** 3 – 6 / 1000 live births. (≃1%in UK)
- **Age:** Between 2 – 4 years of age.
- **Sex:** more common in boys. (4 time more)
- **Genetic factors:** Risk in siblings
- Monozygotic twin concordance 60%
- More common in industrial areas.
- **The prevalence:** in the last few years.

Clinical pictures:

Time: features must be pervasive, i.e. present in all settings and cause functional impairment.

Onset: any age, but often in the preschool years between 2 and 4 years (<3yrs).

A–Social impairment:

- Lack of social smiling
- Difficulty making friends, lack of interest in peers.
- Poor understanding of gestures, impaired imagination. ,
- Abnormal eye contact, body language
- Fail to point–fail to imitate–talk at rather than with people– mind blind.

B–Abnormal speech & language development (Delay In speech)

C–Restricted repetitive patterns of behavior interests or activities:

1–Stereotyped or repetitive movements, use of objects or speech

- Stereotyped behavior, repetitive play (arm flapping, grimacing, hand, tiptoe walking with stiffening and posturing)
- Echolalia (repetition of words, phrases, intonation, or sounds of the speech of others)
- Neologisms: use of jargon or invented words with no meaning .
- Poor concentration span–Ordering same toys

2– Adherence to routines or ritualized patterns of behavior (verbal or nonverbal) Marked resistance to any change to daily route

3–Highly restricted interests (abnormal in strength or focus) e.g. Preoccupation with certain objects

4–Unusual response to sensory aspects of the environment e.g.

- Adverse response to particular sounds (covering ears to certain sounds)
- Apparent indifference to temperature
- Excessive touching/smelling of objects

AUTISM ASSOCIATED FEATURES

- Mental retardation (Distractibility–Poor organizational ability–Difficulty with abstraction).
- Affective disorders – anxiety, sleep disturbance.
- Seizures affect in 25% of cases (onset 11–14 years).
- Diminished response to pain & external stimuli & self–injury 10 – 15% (Head banging – biting of tongue – scratching of skin)
- General learning and attention difficulties (about two–thirds)
- Gastrointestinal symptoms, eating problems

Diagnosis: (DSM IV criteria) Diagnostic and Statistical Manual of Mental Disorders: Six or more items from all three areas are required to make the diagnosis must be present prior to 3 years of age.

- Detailed medical/developmental history.
- Family history.
- Examination:(dysmorphology–OFC).
- Audiological work–up.
- Screening tools (DSM–5 –M–CHAT)
- EEG (Epilepsy).
- Karyotyping (if you suspect syndrome).
- Neuroimaging MRI CT BRAIN (not routine).
- Metabolic screen (+ve F/H–progreesive).

Management: multidisciplinary team

- Parental education and support.
- Early detection & intervention
- based special education (Speech therapy –Occupational therapy)
- Behavioral therapies (ABA =Applied behavior analysis)
- Community service. (Financial support–Special schools and education)

Differential diagnosis:

- Asperger syndrome: Like autism but milder– No delay in language –Vocabulary & grammar development normal more social activity & more communication.
- Childhood disintegrative disorders. (Rapid deterioration)
- Rett syndrome (XLD –occurs almost exclusively in females)

- Mental retardation, hearing impingement.
- Childhood schizophrenia.

Note: early detection and intervention–Language skills & IQ are the best indicators of long–term function.

Attention deficit hyperactivity disorder (ADHD)

- ADHD is the **most common** neurobehavioral disorder of childhood.
- 7–10% of school–age children meet diagnostic criteria for ADHD (more common in **boys**).
- Affected children have problems in **3 areas: inattention** صعوبة التركيز – **hyperactivity** نشاط مفرط **and impulsiveness** اندفاعي **acting before thinking** (suggestive symptoms should be present for > **6 months**).

- Etiology: **Multifactorial** (Genetic predisposition–environmental as fetal exposure to smoking or alcohol– abnormalities in neurotransmitters)

- Risk Factors:
  - Male > Female 4:1.
  - Learning difficulties, mental health disorders, autism.
  - CNS disorders e.g epilepsy, C.P.
  - 1st degree relative with ADHD.
  - Extremely premature infants, & VLBW.
  -

DSM–5 Diagnostic Criteria for ADHD:

≥ 6 of the following symptoms of INATTENTION	≥ 6 of the following symptoms of HYPERACTIVITY–IMPULSIVITY.	Other Diagnostic Criteria
1. Failure to give attention to details with making careless mistakes in school work. 2. Fallure to finish schoolwork or follow instructions. 3. Failure to sustain attention to tasks or play activity. 4. Failure to organize activities. 5. Avoidance to engage to tasks that needs sustained mental effort. 6. Frequent loss of necessary things as pencils or books. 7. Easily distracted. 8. Forgetfulness. 9. Does not seem to listen when spoken to directly	Hyperactivity: 1. Move around nervously with hands and feet twisting in seat. 2. Unable to sit still in classroom with frequent walking around. 3. Talks excessively. 4. Excessive inappropriate running. 5. Difficulty in playing quietly. 6. Acts as if (driven by a motor) Impulsivity: 1. Answering before the question is completed. 2. Interrupting others. 3. Cannot wait his turn.	1. Symptoms present before age of 12 years. (Old: was <6–7years) 2. Symptoms noted at least in 2 different settings (i.e. home or school). 3. Symptoms causing considerable school and social & functional impairment. 4. Behavior should be inconsistent with the child's developmental age. 5. Exclusion of other causes of school underachievement or other psychiatric illnesses.

Management of ADHD:

1. Educational measures (Communication between home, school and clinic. – Special educational provision: One–one teaching & avoid anything which might distract the child)

2. Behavioural management techniques

3. Medication:
   - A–Psychostimulant.
     - **Methylphenidate** (**Ritalin**) : it inhibits the reuptake of dopamine and norepinephrine,& increased dopaminergic and noradrenergic activity in the prefrontal cortex.
       Other drugs (Dexamphetamine–Tricyclic Antidepressant–Risperidone)
   - B–Atomoxetine: selective norepinephrine reuptake inhibitors.

4. Dietary advice: exclusion of certain foods from diet such as red food, cows milk.

Child with ADHD has an increased risk of:

Low self–esteem, poor peer relations, poor mental health ,anxiety ,and poor school performance & exclusion from schooling and antisocial behavior

Learning or intellectual disability: is the medical term used when school–age children have significantly reduced ability to understand new or complex information and to learn new skills, and reduced ability to cope independently with everyday life

The degree of LD is classified to mild, moderate or severe based on functional assessment of communication or intelligence quotients (IQ).

Specific learning difficulties

These have an impact on a specific area of the child or young person's life/skills.
These include:

- **dyslexia** – difficulties with reading .
- **dyscalculia** – difficulties in learning and comprehending numbers .
- **dysgraphia** – difficulties with writing skills.

Neurocutaneous syndromes.

The nervous system & skin have a common **ectodermal** origin.
Embryological disruption causes syndromes involving abnormalities of both Systems.

Neurocutaneous syndromes Includes:

1. Neurofibromatosis.
2. Tuberous sclerosis.
3. Sturge – Weber syndrome.
4. Ataxia telangiectasia.
5. Von Hippel – Lindau disease.

1–Neurofibromatosis (NF)

Autosomal dominant disorder (AD) (30–50 % new mutation)

Types:

1. **NF type I (90%): NF1 gene defect in Ch 17**
2. **NF type II (10%): NF 2 gene defect in Ch 22**

Type I Neurofibromatosis (mutation in the neurofibromin–1 (NF1) gene)

Clinical features: To make the diagnosis, **two or more** of these **criteria** need to be present:

1. Six or more **cafe–au–lait spots** greater than 5 mm in size before puberty, greater than 15 mm after puberty
2. Two or more **neurofibroma**, a firm nodular overgrowth of any peripheral nerve
3. **Axillary freckling**
4. **Optic glioma** which may cause visual impairment
5. One or more **Lisch nodule**: a hamartoma of the iris seen on slit–lamp examination
6. **Bony lesions** from sphenoid dysplasia, which can cause eye protrusion
7. **First–degree** relative with NF–1.



Complications of NF I:

- Risk of benign & malignant tumors.
- Hypertension associated with pheochromocytoma ،renal artery stenosis
- Macrocephaly, Megalencephaly
- Visual or auditory impairment Learning difficulties, epilepsy
- Scoliosis.
- Both NF–1 and NF–2 can be associated with endocrine disorders, the multiple endocrine neoplasia syndromes.

NF type II: (mutation in the NF2 gene) less common than type I.

- Bilateral acoustic neuromata are the predominant feature
- Associated with inherited schwannomas, meningiomas, and ependymomas.
- Cerebello– pontine angle syndrome with a facial (VIIth) nerve paresis and cerebellar ataxia.

2–Tuberous sclerosis (TSC)

TSC is AD disorders (**70% new mutation**)
Prevalence: 1:9000 Live births.
Clinical picture:

Skin (70 – 80%)	Nervous system (≃ 50%)
• **Depigmented (Ash leaf) Patches.**  FIGURE 2: "Ash-Leaf" hypomelanotic macule • Shagreen patch usually over the lumbar spine • Adenoma sebaceum usually over the nose and cheeks (>3yrs) 	• Epilepsy (Infantile spasm– autism – ADHD) • Intellectual disability & MR. • Astrocytoma –hydrocephalus. • MRI & CT brain: Shows calcification. **Other systems involved:** **CVS:** Cardiac rhabdomyomas (echocardiography) **Renal:** angiomyolipomas and polycystic kidneys. **Eyes:** Retinal astrocytomas. **Lungs:** cysts **Retina:** (phakomata) from local degeneration

3-Sturge weber syndrome



This is a sporadic disorder characterized by:

- **Port wine stain:** Hemangioma over the ophthalmic division of trigeminal nerve.
- Epilepsy, MR.
- Eye: congenital glaucoma 50% of cases.
- Contralateral hemiplegia – learning difficulties.
- CT or MRI brain shows tramline calcification (ipsilateral leptomeningeal angioma).

DD of cafe au lait spots	DD of intracranial calcification
(1). Neurofibromatosis.NF (2). Tuberous sclerosis. (3). Ataxia Telangiectasia. (4). Noonan's syndrome. (5). Russel silver syndrome. (6). Mccune Albright syndrome.	(1). NF – Tuberous sclerosis (2). Intracranial Hge. (3). TORCH. (4). Craniopharyngioma. (5). Vit D excess (6). Hypo / Hyperparathyroidism.

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2- Illustrated textbook of paediatrics 6th edition.
3- Nelson essentials of pediatrics 9th edition.
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تنبيه:

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